

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv('online_shoppers_intention.csv')
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Administrative	Administrative_Duration	Informational	Informational_Duration	ProductRelated	Prc
0	0	0.0	0	0.0	1	
1	0	0.0	0	0.0	2	
2	0	0.0	0	0.0	1	
3	0	0.0	0	0.0	2	
4	0	0.0	0	0.0	10	



```
In [4]: df.shape
```

```
Out[4]: (12330, 18)
```

```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12330 entries, 0 to 12329
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Administrative                        12330 non-null  int64
1   Administrative_Duration               12330 non-null  float64
2   Informational                         12330 non-null  int64
3   Informational_Duration                12330 non-null  float64
4   ProductRelated                       12330 non-null  int64
5   ProductRelated_Duration              12330 non-null  float64
6   BounceRates                          12330 non-null  float64
7   ExitRates                            12330 non-null  float64
8   PageValues                           12330 non-null  float64
9   SpecialDay                           12330 non-null  float64
10  Month                                12330 non-null  object
11  OperatingSystems                     12330 non-null  int64
12  Browser                              12330 non-null  int64
13  Region                              12330 non-null  int64
14  TrafficType                          12330 non-null  int64
15  VisitorType                           12330 non-null  object
16  Weekend                              12330 non-null  bool
17  Revenue                              12330 non-null  bool
dtypes: bool(2), float64(7), int64(7), object(2)
memory usage: 1.5+ MB
```

```
In [6]: df.describe()
```

Out[6]:	Administrative	Administrative_Duration	Informational	Informational_Duration	ProductRelated
count	12330.000000	12330.000000	12330.000000	12330.000000	12330.000000
mean	2.315166	80.818611	0.503569	34.472398	31.731468
std	3.321784	176.779107	1.270156	140.749294	44.475503
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	7.000000
50%	1.000000	7.500000	0.000000	0.000000	18.000000
75%	4.000000	93.256250	0.000000	0.000000	38.000000
max	27.000000	3398.750000	24.000000	2549.375000	705.000000



In [7]: `df.isnull().sum()`

```
Out[7]: Administrative      0
Administrative_Duration    0
Informational              0
Informational_Duration      0
ProductRelated             0
ProductRelated_Duration    0
BounceRates               0
ExitRates                 0
PageValues                0
SpecialDay                 0
Month                     0
OperatingSystems           0
Browser                   0
Region                    0
TrafficType               0
VisitorType               0
Weekend                   0
Revenue                   0
dtype: int64
```

There are no null values in the dataset

In [8]: `df.duplicated().sum()`

Out[8]: `np.int64(125)`

There are 125 Duplicated rows in the dataset

```
In [9]: df.columns = df.columns.str.lower()
df.columns = df.columns.str.strip()
df.columns = df.columns.str.replace(' ', '_')
```

In [10]: `df.head()`

Out[10]:	administrative	administrative_duration	informational	informational_duration	productrelated	prod
0	0	0.0	0	0.0	1	
1	0	0.0	0	0.0	2	
2	0	0.0	0	0.0	1	
3	0	0.0	0	0.0	2	
4	0	0.0	0	0.0	10	

Handling Duplicate Values

Checking the percentage of duplicates in the dataset

```
In [11]: percentage = df.duplicated().sum() / len(df)*100
print('The Percentage of Duplicates is the dataset is : ', percentage)
```

The Percentage of Duplicates is the dataset is : 1.013787510137875

Since the Duplicates consist of only 1.01% of the total dataset , we would proceed to drop these duplicates

```
In [12]: df = df.drop_duplicates()
```

```
In [13]: print("Shape after removing duplicates:", df.shape)
print("Duplicates:", df.duplicated().sum())
```

Shape after removing duplicates: (12205, 18)

Duplicates: 0

Now There are no Duplicates In The Dataset

Target Variable Analysis

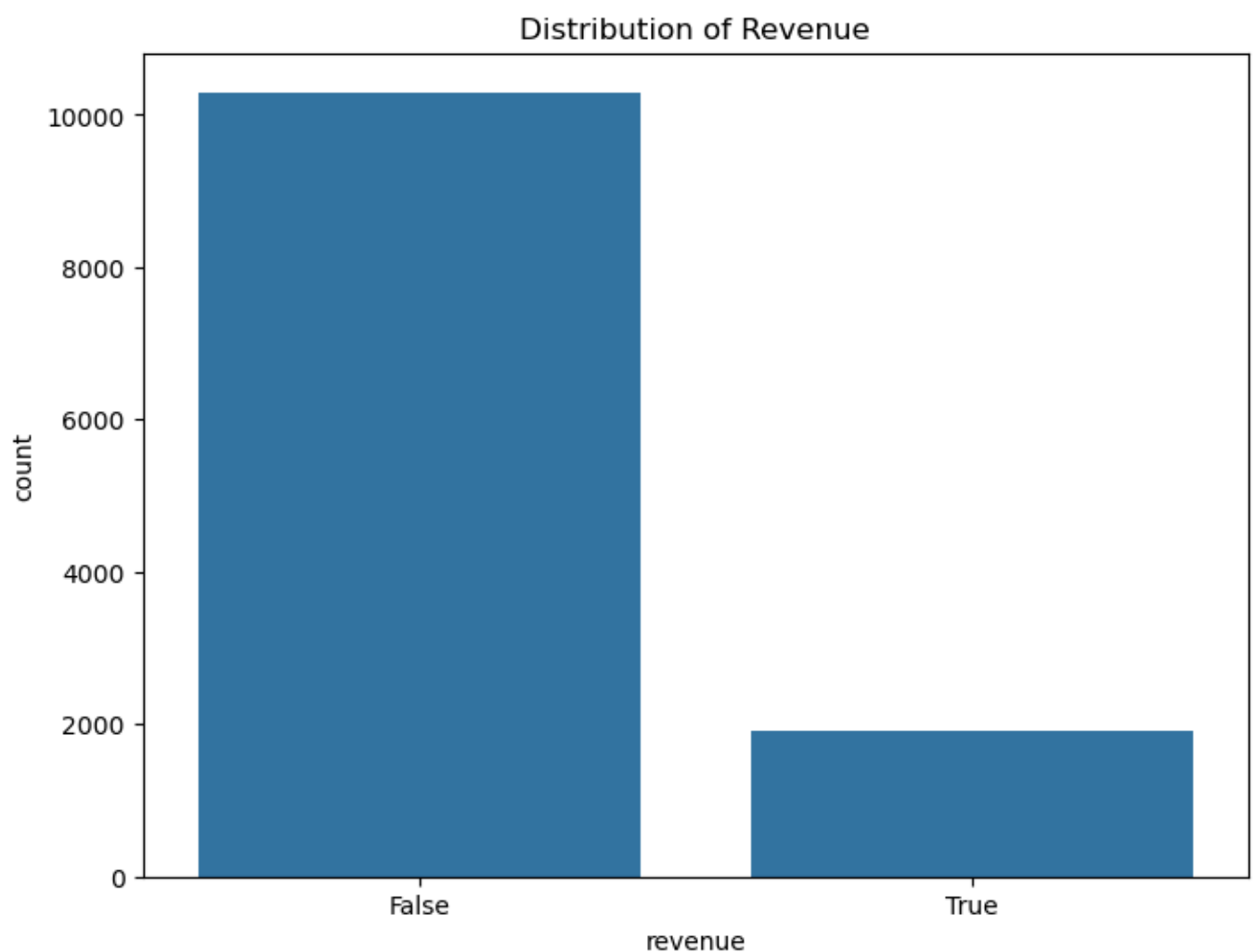
```
In [14]: df['revenue'].value_counts()
```

```
Out[14]: revenue
False    10297
True      1908
Name: count, dtype: int64
```

```
In [15]: (df['revenue'].value_counts(normalize=True)*100)
```

```
Out[15]: revenue
False    84.367063
True     15.632937
Name: proportion, dtype: float64
```

```
In [16]: plt.figure(figsize=(8,6))
sns.countplot(
    data= df,
    x='revenue'
)
plt.title('Distribution of Revenue')
plt.show()
```



Observations:

1. We can observe that only 15% of the customers bring revenue
2. The dataset is also imbalanced

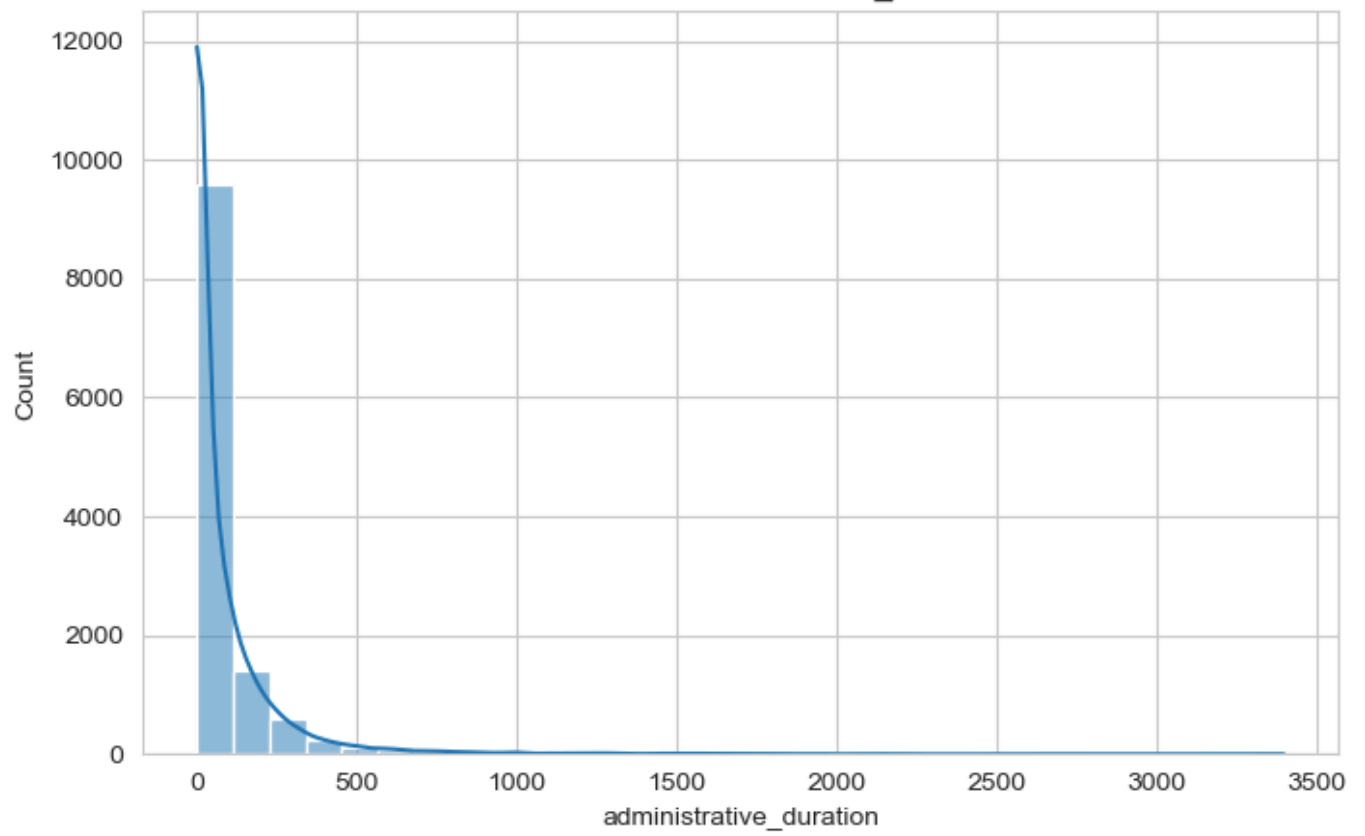
Seperating Numerical and Categorical Columns

EDA

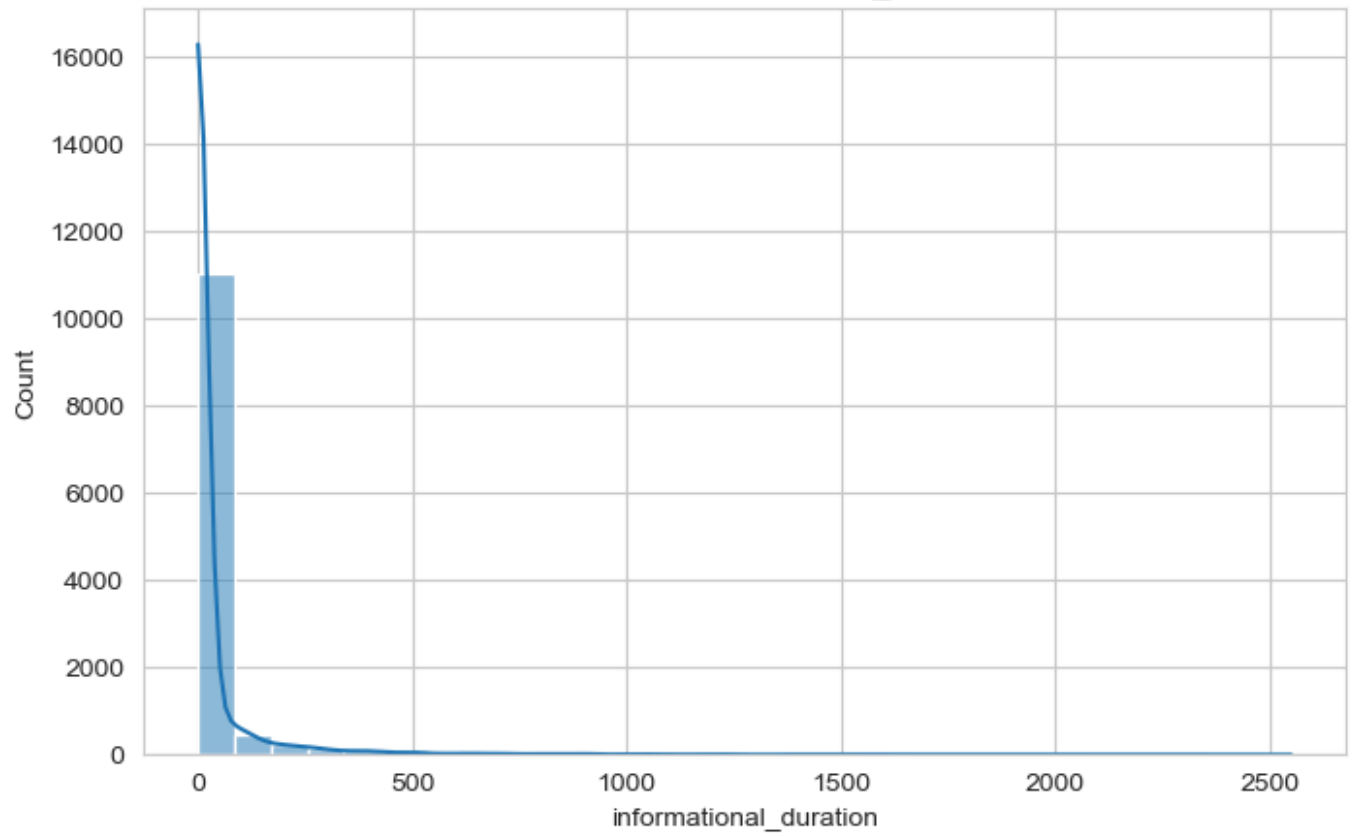
```
In [17]: continuous_cols = [  
    'administrative_duration',  
    'informational_duration',  
    'productrelated_duration',  
    'bouncerrates',  
    'exitrates',  
    'pagevalues',  
    'specialday'  
]
```

```
In [18]: sns.set_style('whitegrid')  
  
for col in continuous_cols:  
    plt.figure(figsize=(8,5))  
    sns.histplot(data=df, x=col, bins=30, kde=True)  
  
    plt.title(f'Distribution of {col}')  
    plt.show()
```

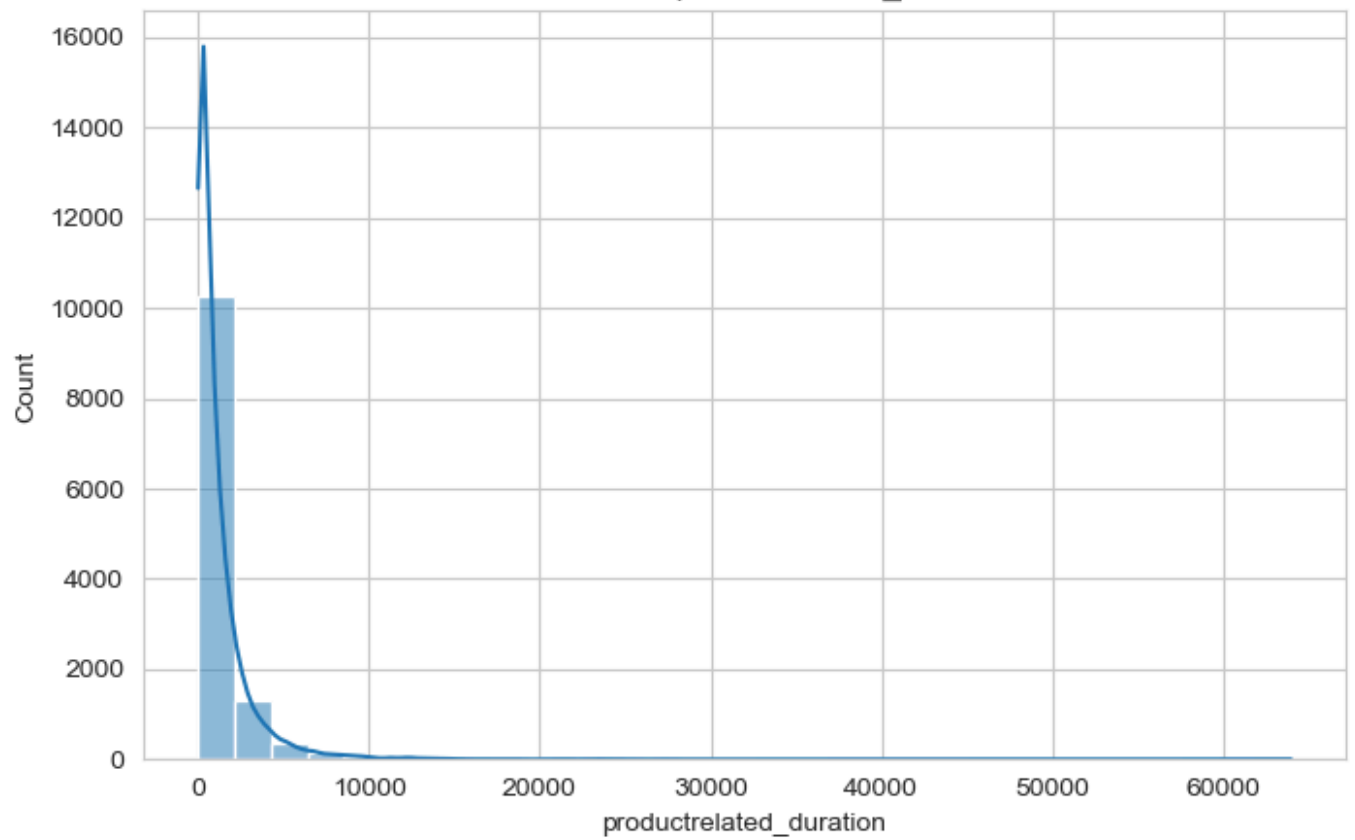
Distribution of administrative_duration



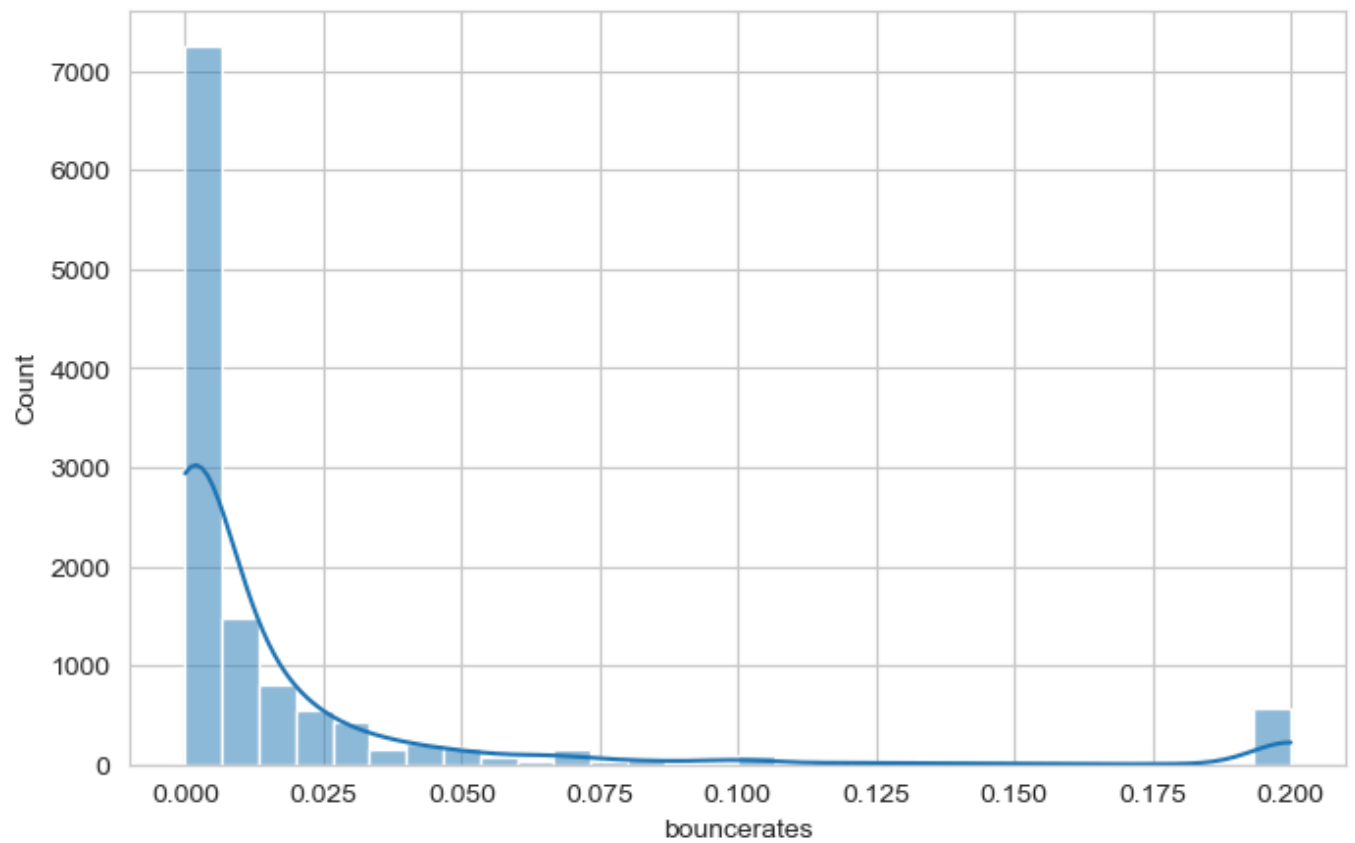
Distribution of informational_duration



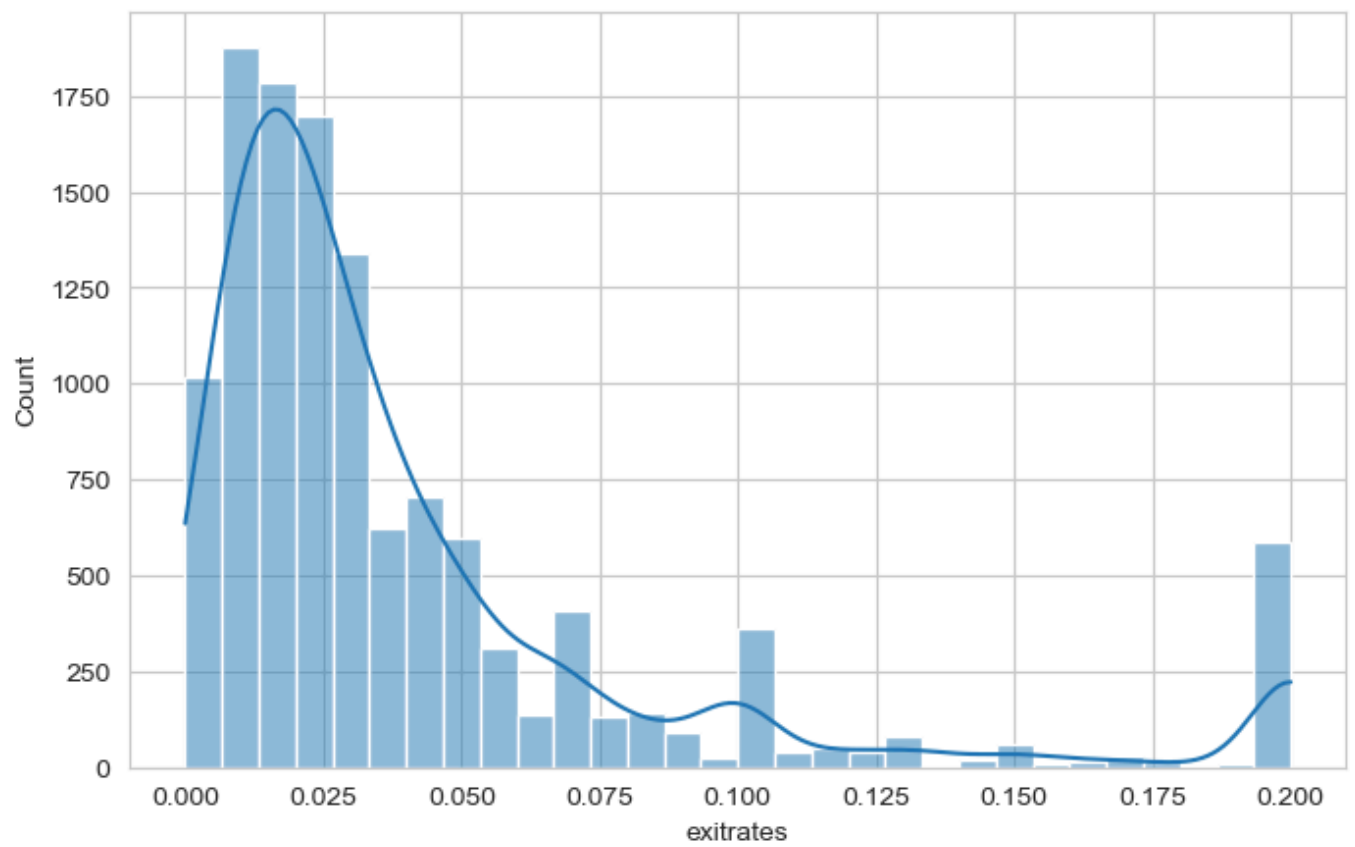
Distribution of productrelated_duration



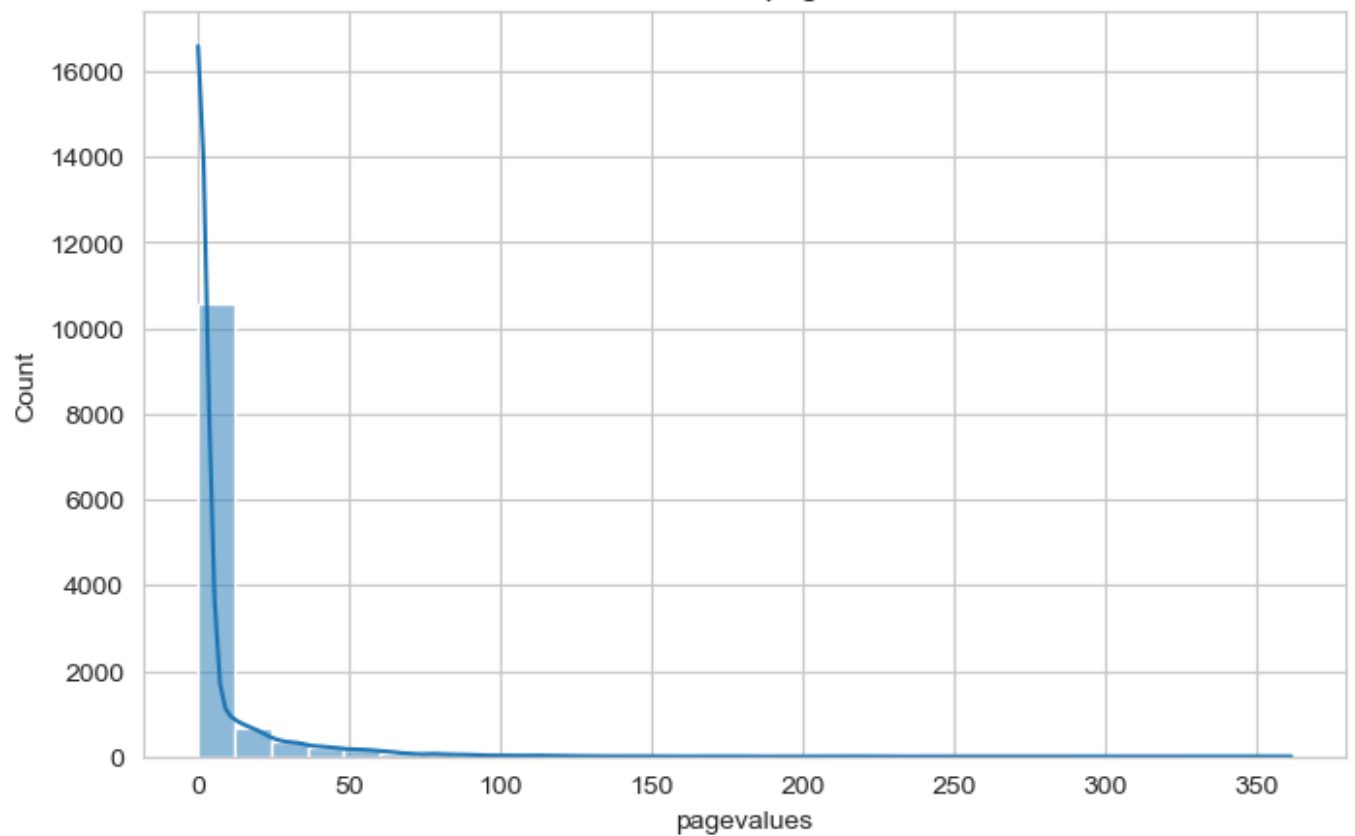
Distribution of bouncerrates

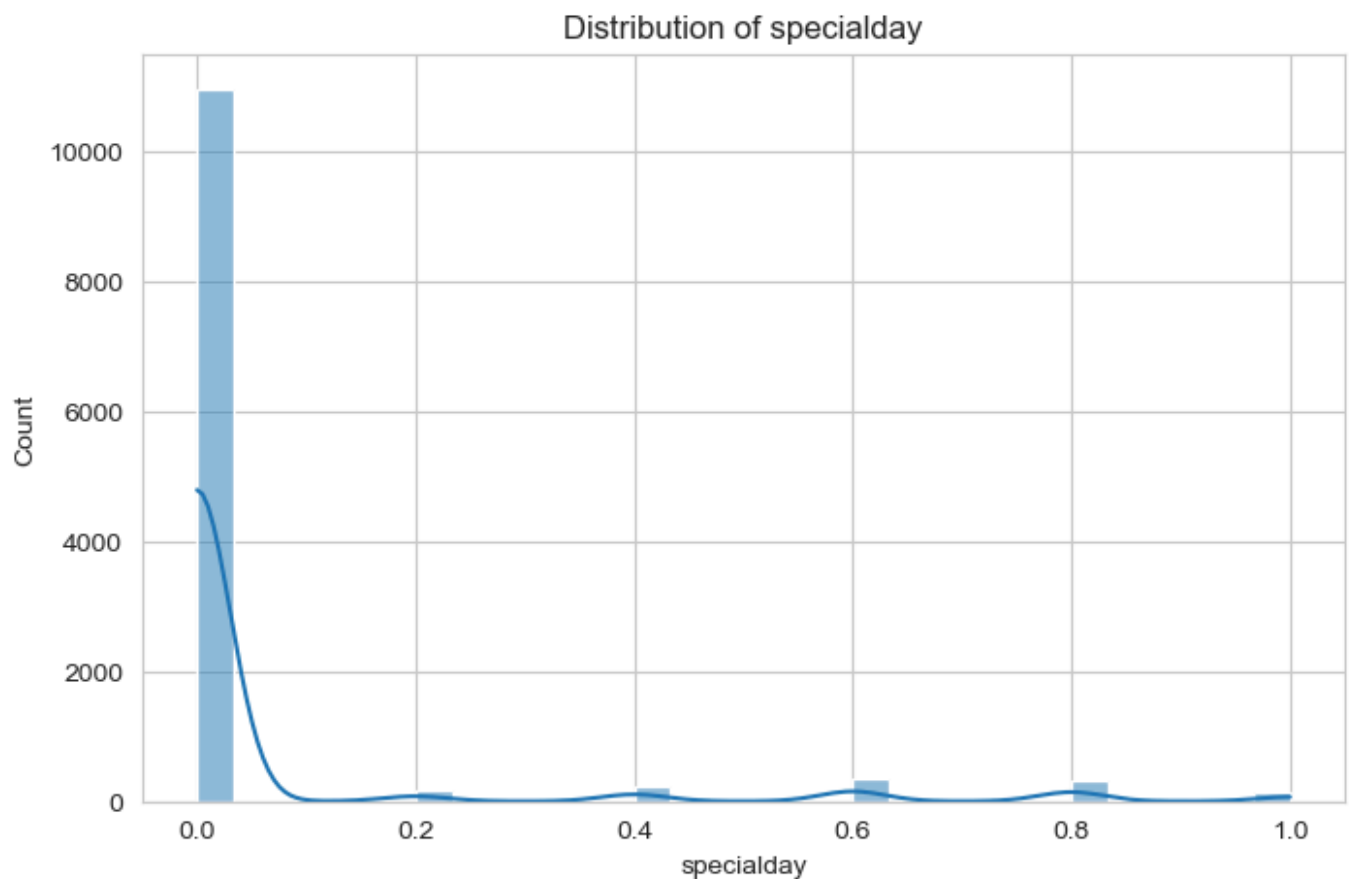


Distribution of exitrates



Distribution of pagevalues





1. Administrative_Duration Observation:

- A. The distribution is highly right-skewed.
- B. Most visitors spend very little time on administrative pages.
- C. A small number of users spend significantly longer durations, creating a long right tail.
- D. The presence of extreme values suggests potential outliers, which are acceptable for tree-based models like Random Forest and XGBoost.

2. Informational_Duration Observation:

- A. The distribution is strongly positively skewed.
- B. Most sessions contain little or no time spent on informational pages.
- C. A few visitors spend considerably more time, indicating occasional deep information-seeking behavior.
- D. Several high-duration observations appear as potential outliers.

3. ProductRelated_Duration Observation:

- A. The feature exhibits an extreme positive skew with a long right tail.
- B. Most users spend relatively little time on product-related pages.
- C. A small group of users spends substantially longer browsing products.
- D. These longer browsing sessions may indicate higher purchase intent and could be important predictors of revenue.

4. BounceRates Observation:

- A. Bounce rates are concentrated near zero, indicating that many users continue browsing beyond the landing page.
- B. The distribution is highly right-skewed.
- C. A noticeable spike exists around 0.20, representing sessions with relatively high bounce rates.
- D. Higher bounce rates may be negatively associated with purchase completion.

5. ExitRates Observation:

- A. Exit rates are moderately right-skewed.
- B. Most observations lie between 0.01 and 0.05.
- C. A secondary concentration near 0.20 suggests some sessions end with relatively high exit probabilities.
- D. Exit rate is expected to be an important predictor of purchasing behavior.

6. PageValues Observation:

- A. The majority of observations have PageValues close to zero.
- B. The feature is highly right-skewed with a few very large values.
- C. High PageValues typically indicate pages that contribute significantly to conversions.
- D. This feature is expected to have strong predictive power for the target variable.

7. SpecialDay Observation:

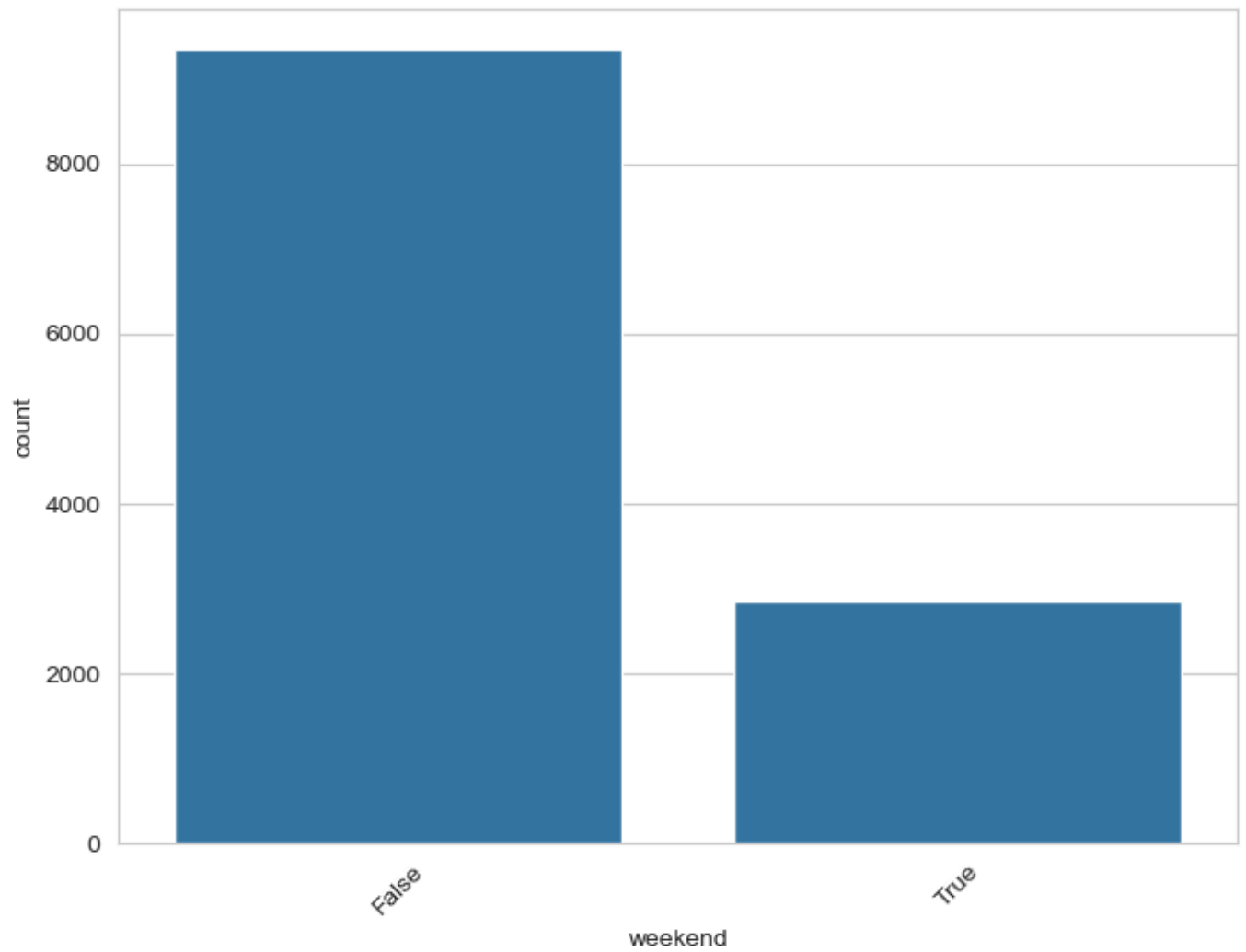
- A. Most sessions occurred when SpecialDay = 0, indicating they were not close to a special shopping event.
- B. Only a small proportion of observations correspond to values between 0.2 and 1.0.
- C. The feature is highly imbalanced and may have limited standalone predictive power, though it could still contribute when combined with other features.

Overall Observation:

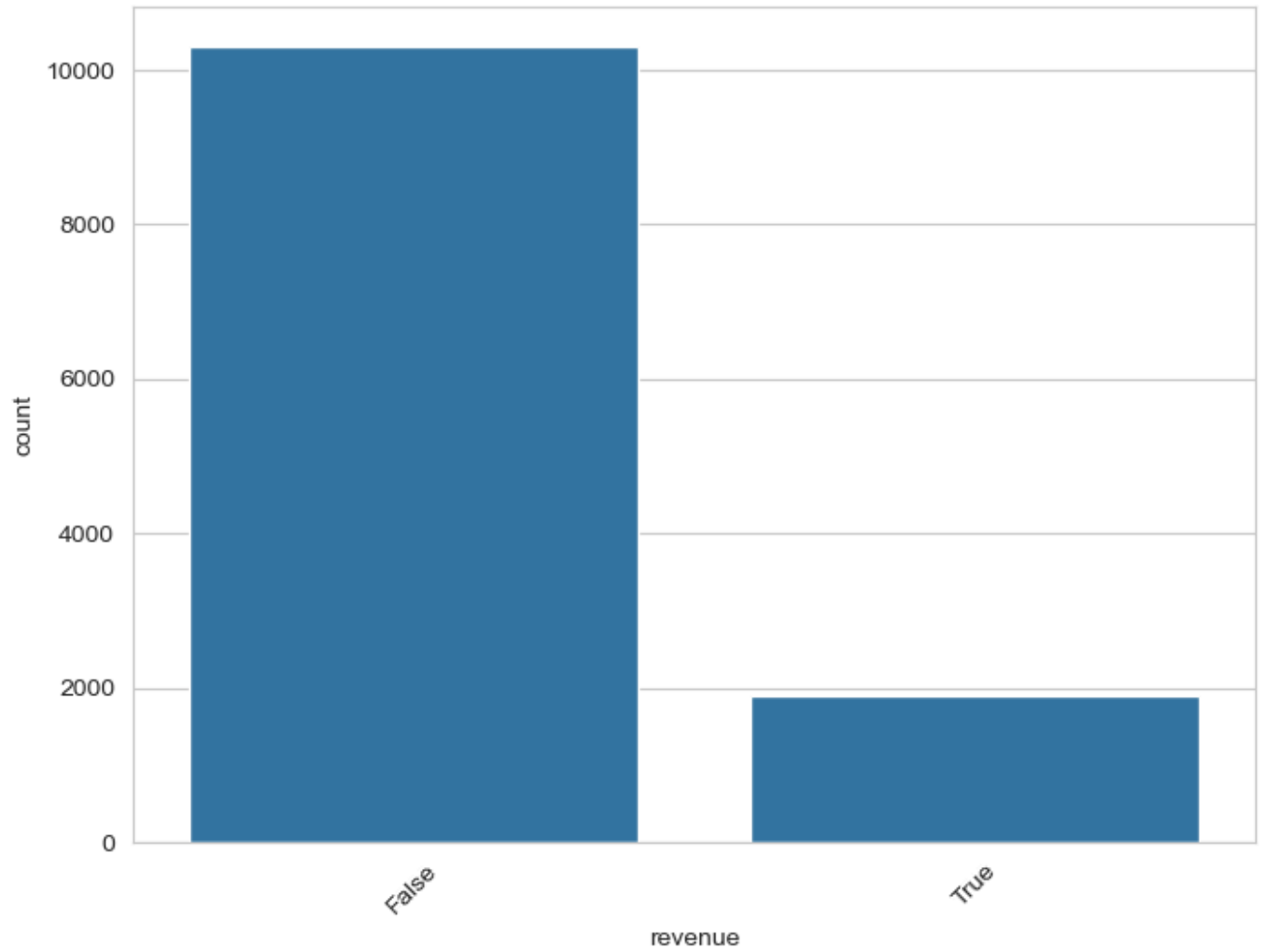
1. Most continuous features exhibit positive (right) skewness, with a large concentration of observations near lower values and a small number of high-value observations.
2. Several variables contain potential outliers, particularly the duration-based features and PageValues. Since tree-based algorithms such as Random Forest and XGBoost are robust to outliers, these values will be retained.
3. Features such as ProductRelated_Duration, PageValues, BounceRates, and ExitRates appear particularly informative and are expected to play an important role in predicting customer purchase intention.

```
In [19]: categorical_columns = ['weekend', 'revenue', 'month', 'visitortype']
for cols in categorical_columns:
    plt.figure(figsize=(8,6))
    sns.countplot(
        data=df,
        x=cols
    )
    plt.title(f'Distribution of {cols}')
    plt.xticks(rotation=45)
    plt.show()
```

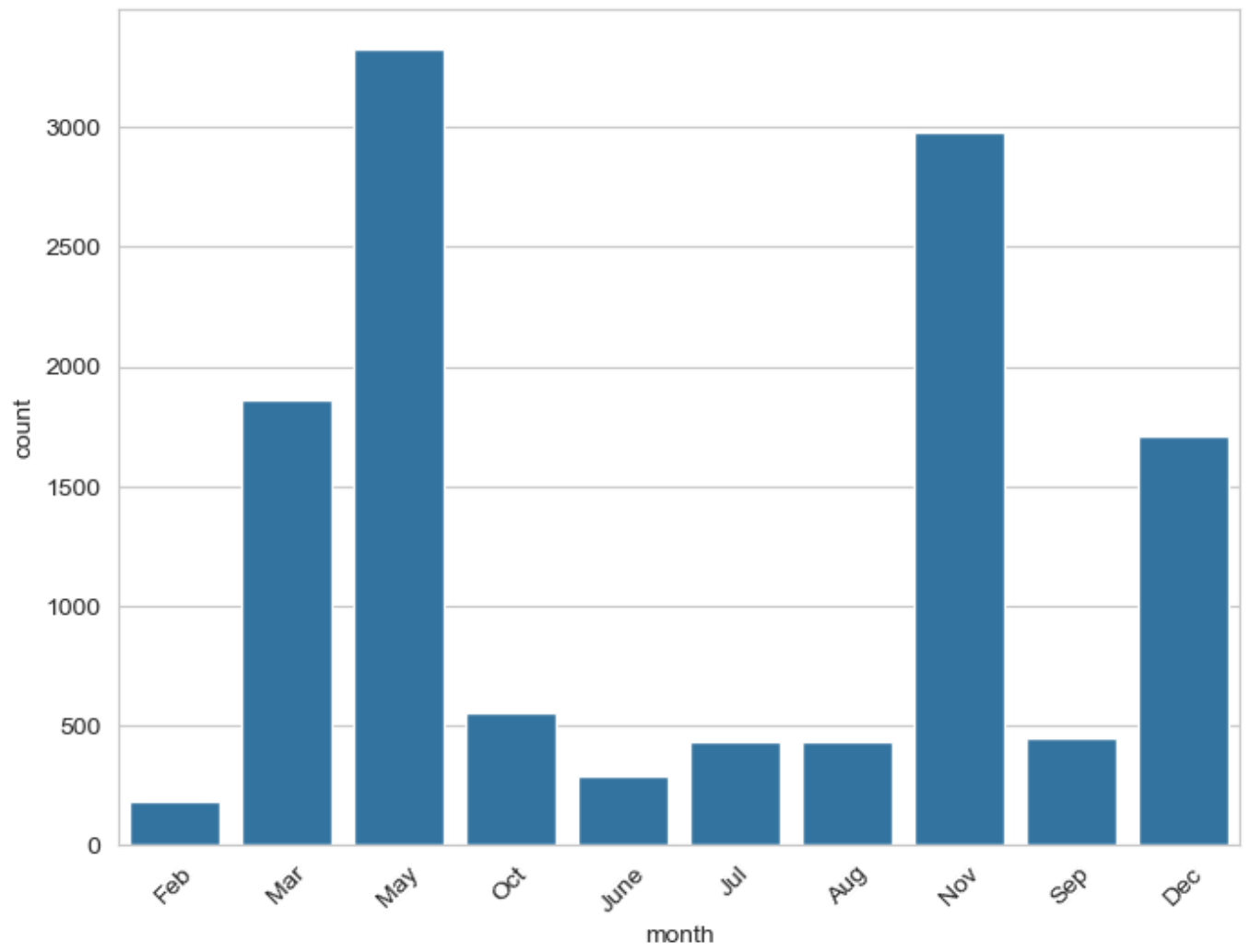
Distribution of weekend

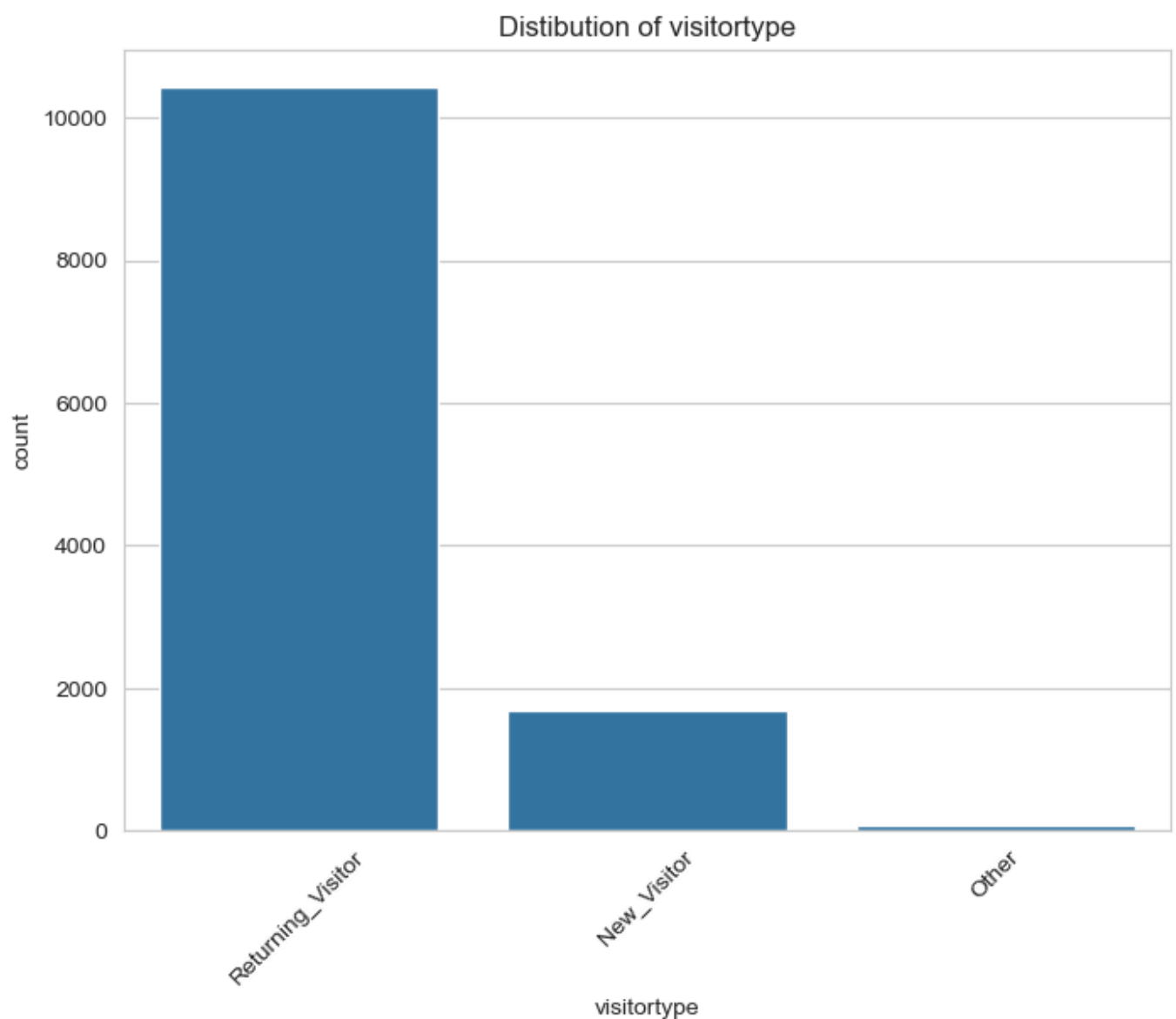


Distribution of revenue



Distribution of month





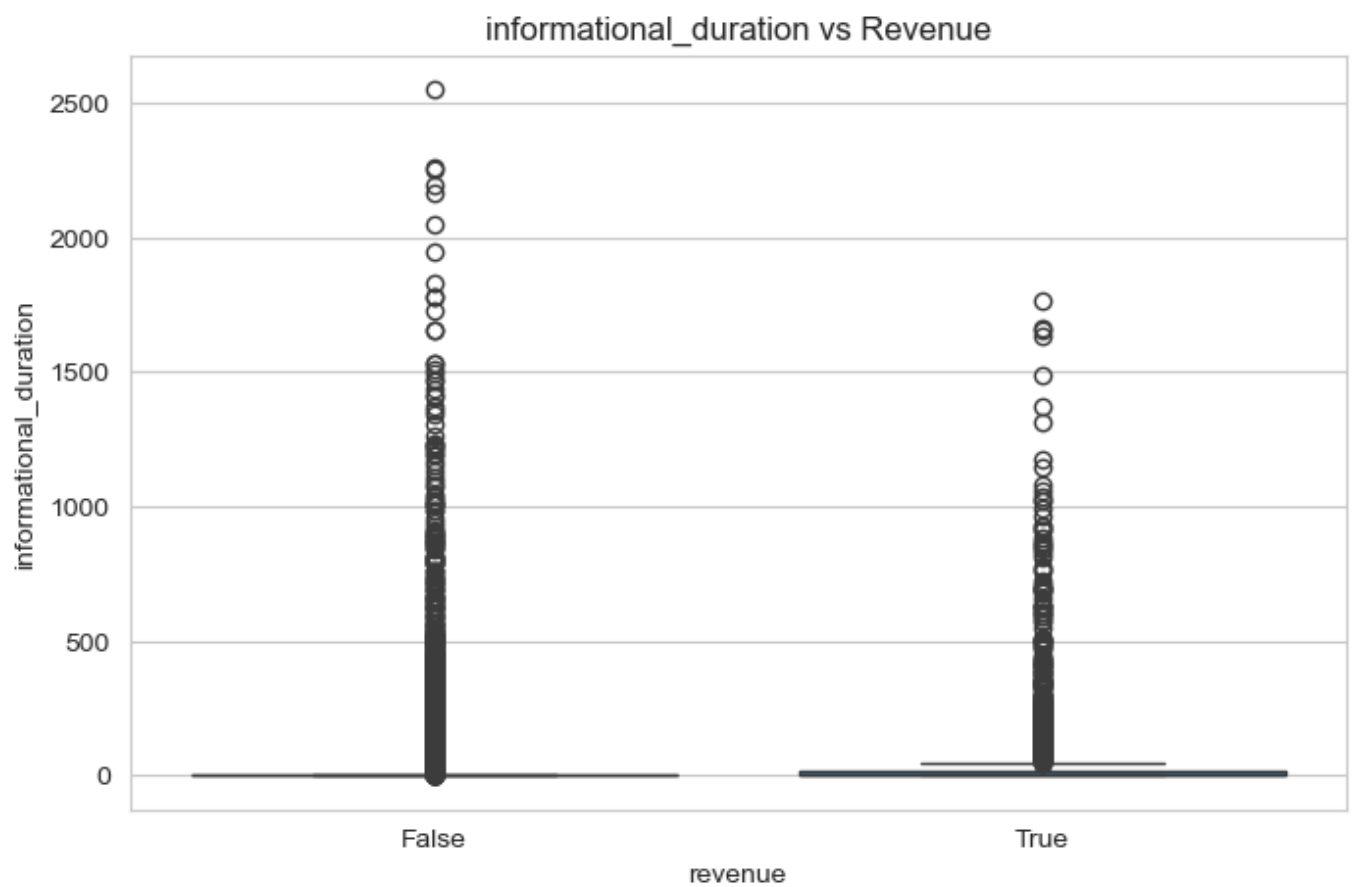
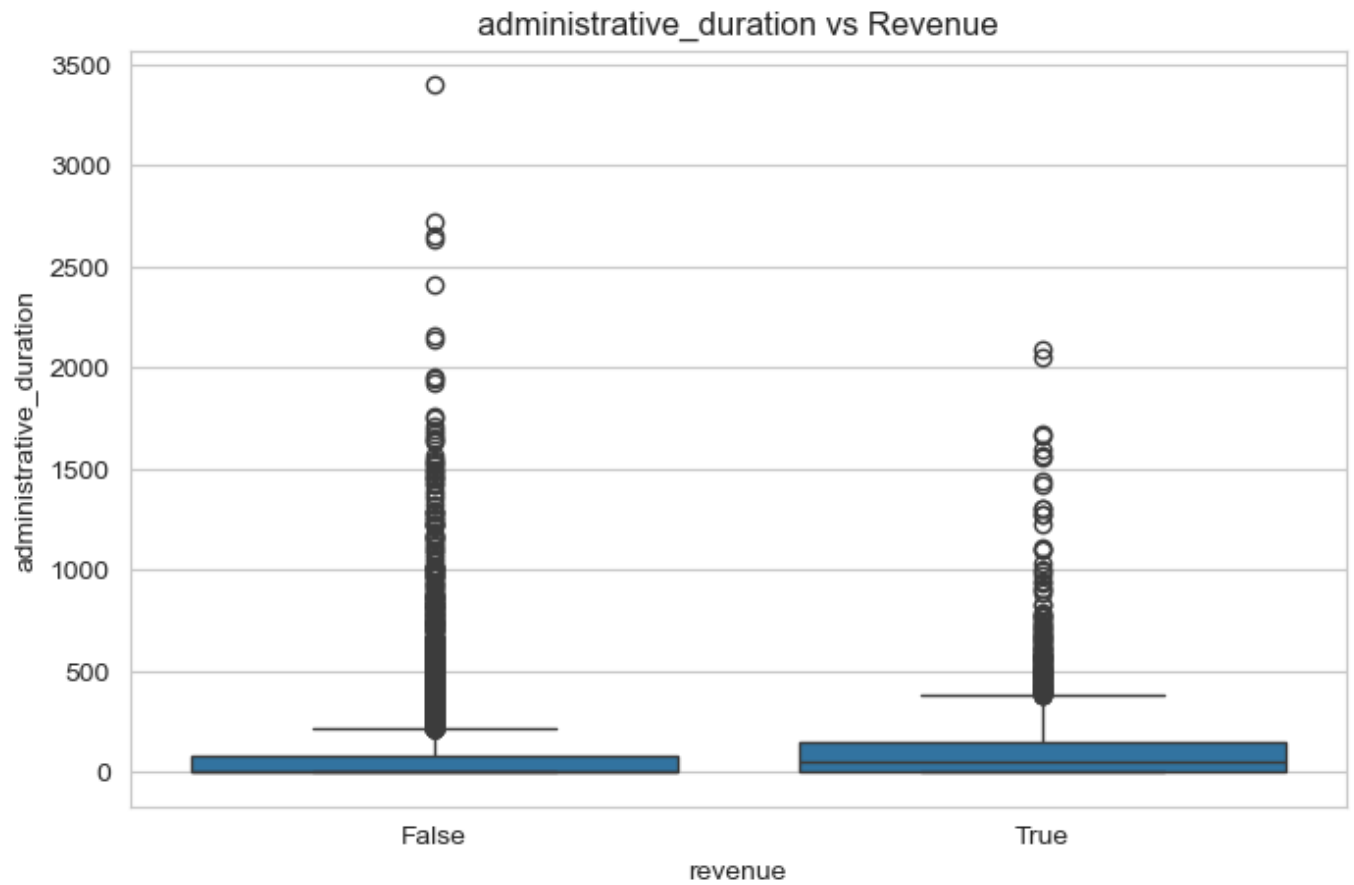
Observations:

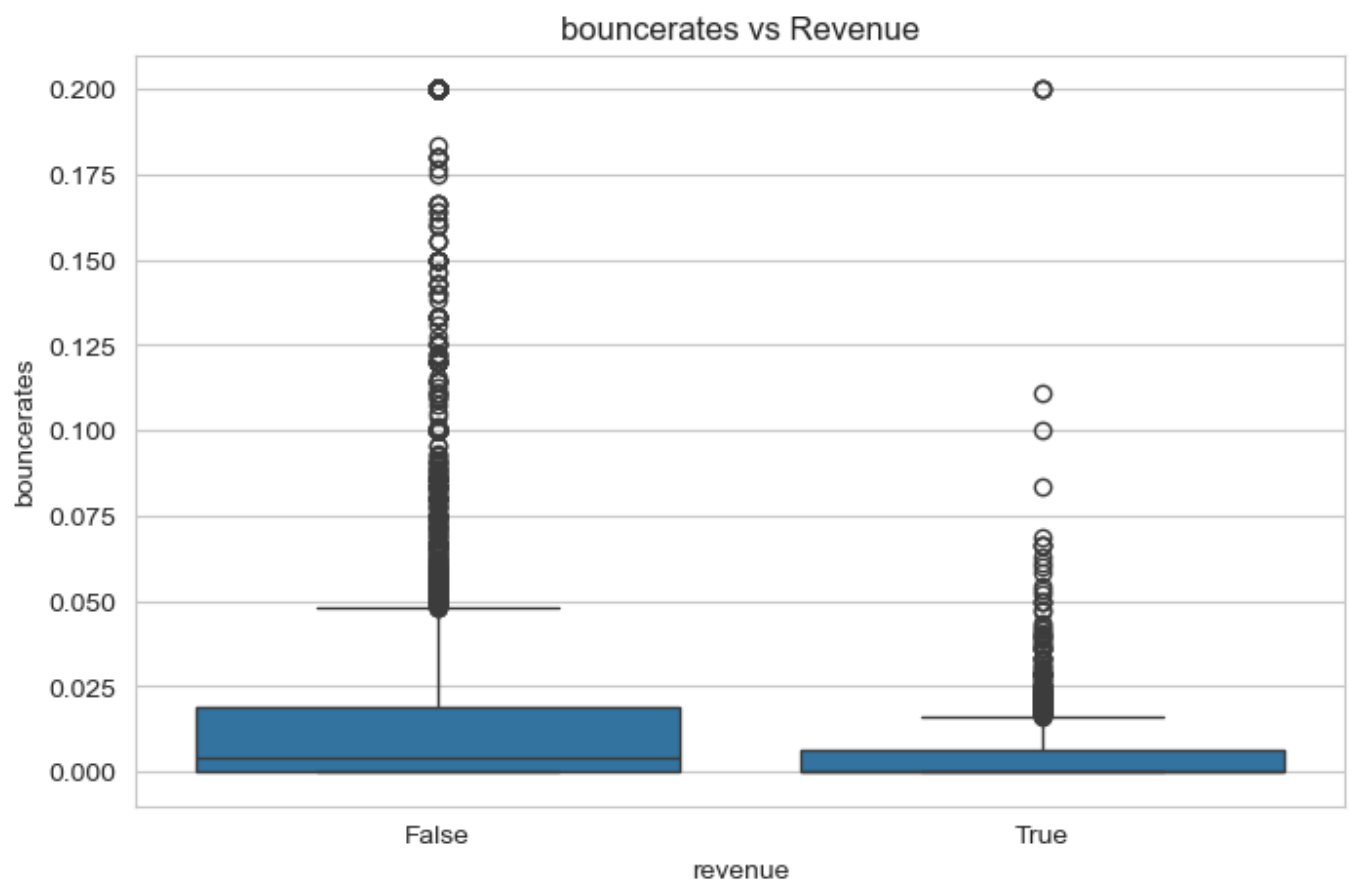
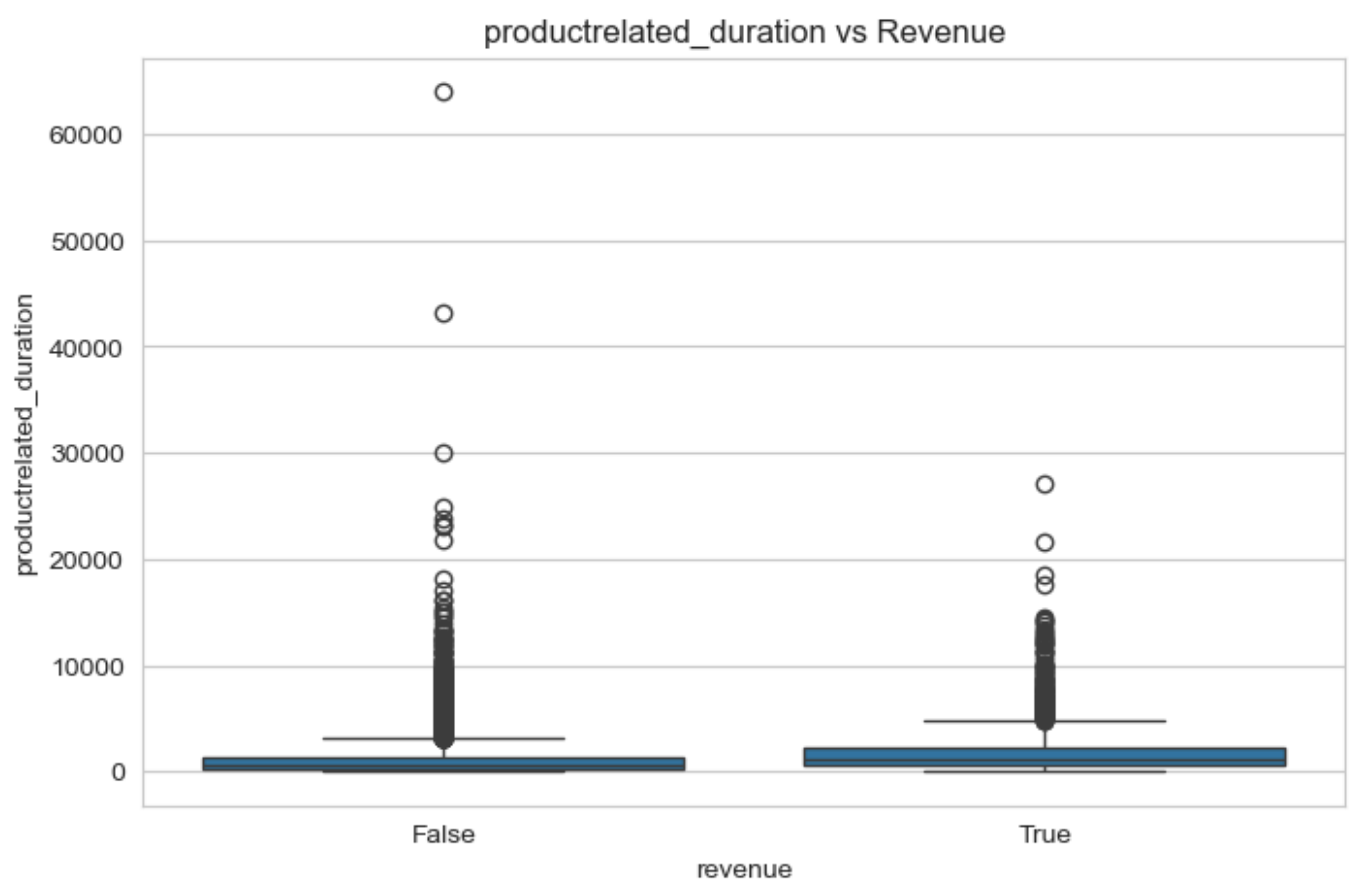
1. Weekend: The majority of website sessions occur on weekdays (Weekend = False), indicating that users are more likely to browse the website during regular working days than on weekends.
2. Revenue: Most website sessions do not result in a purchase (Revenue = False), indicating that only a small proportion of visitors complete a transaction. This also suggests that the target variable is imbalanced.
3. Month: Website traffic is highest during May and November, indicating seasonal variations in customer activity. These months may correspond to promotional events or increased shopping demand.
4. VisitorType: The majority of website visitors are Returning Visitors, suggesting that repeat customers constitute a significant portion of the website traffic and may have higher purchase intent than new visitors.

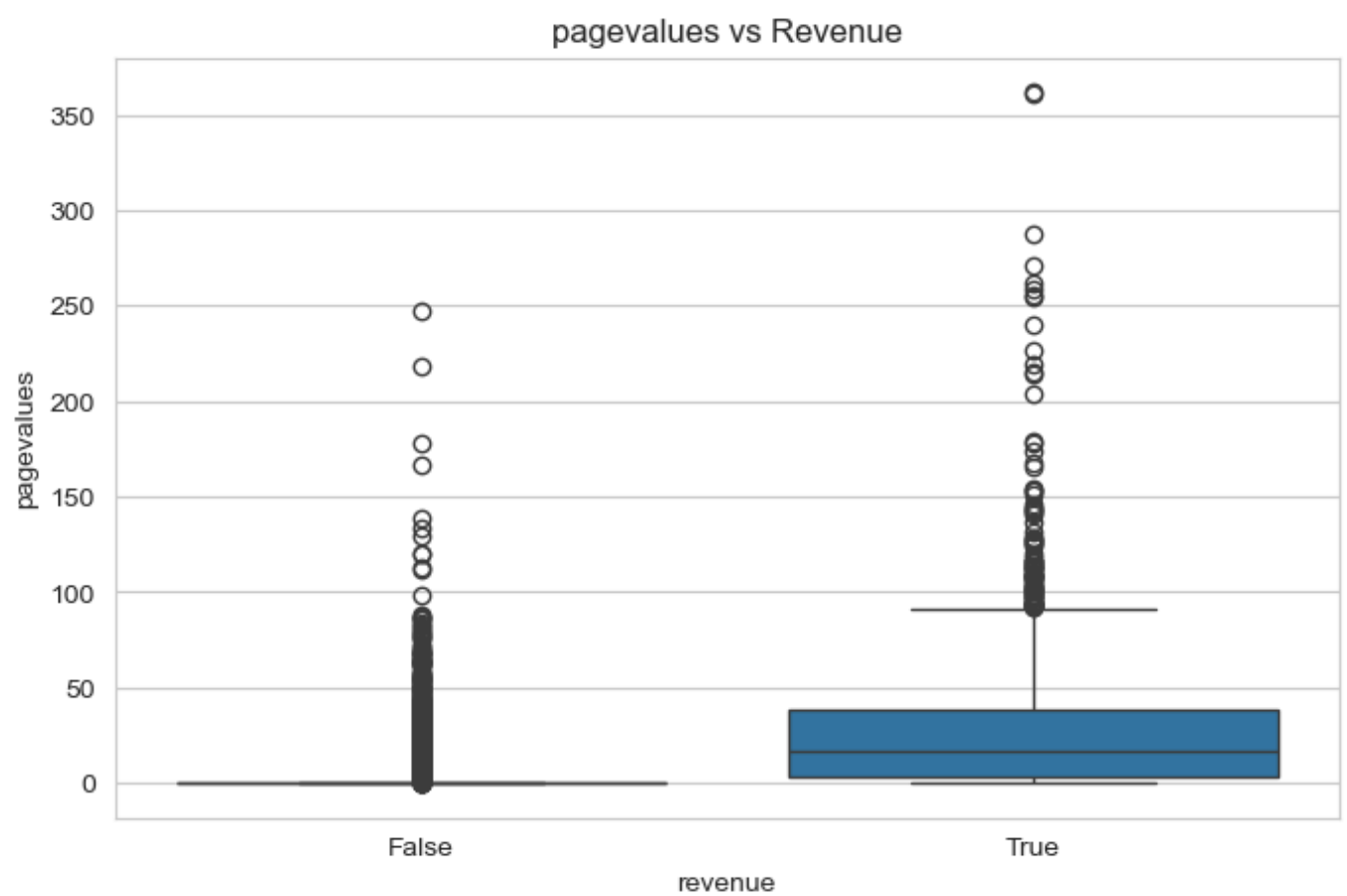
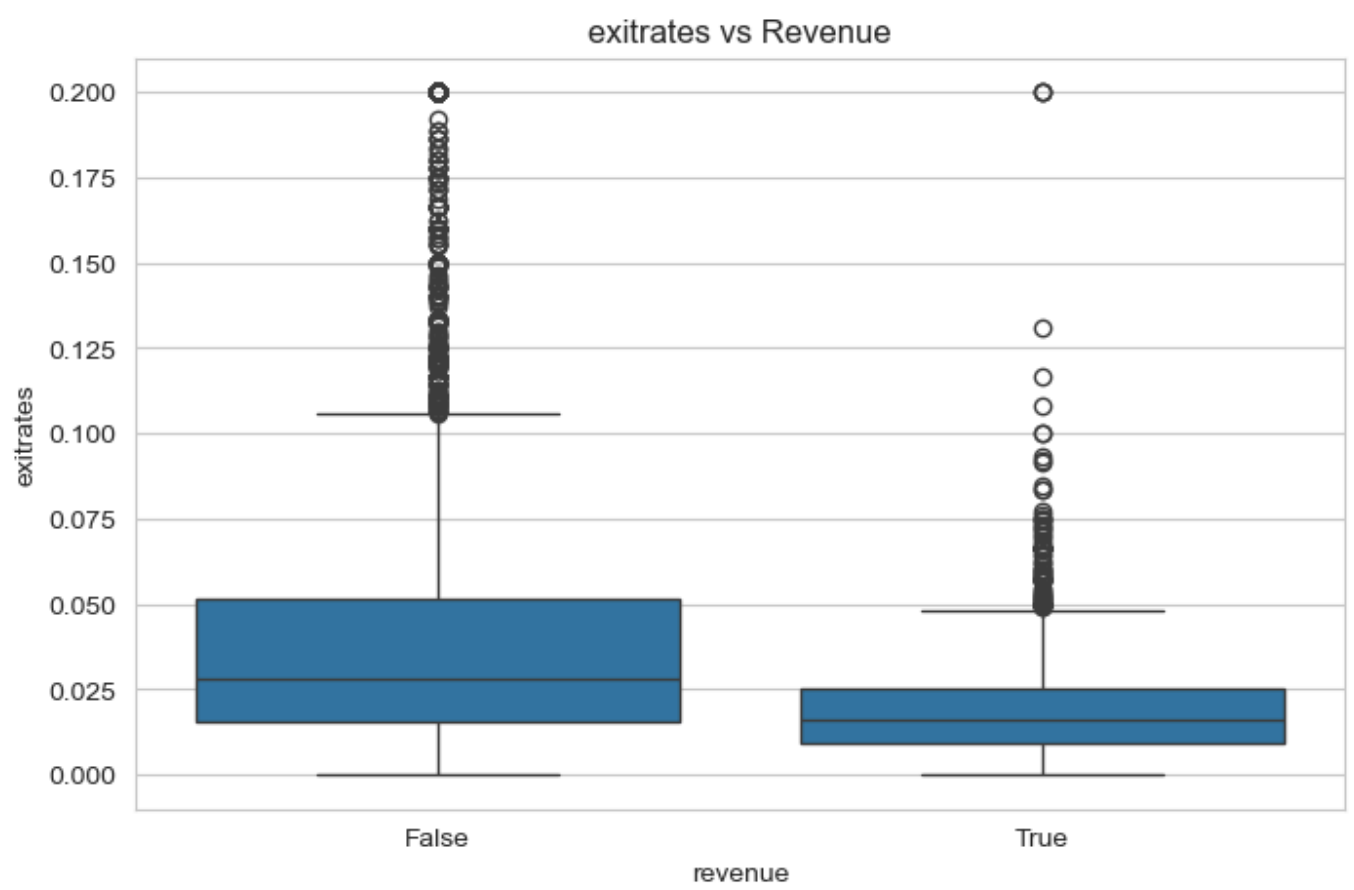
Bivariate Analysis : Continous Features vs Revenue

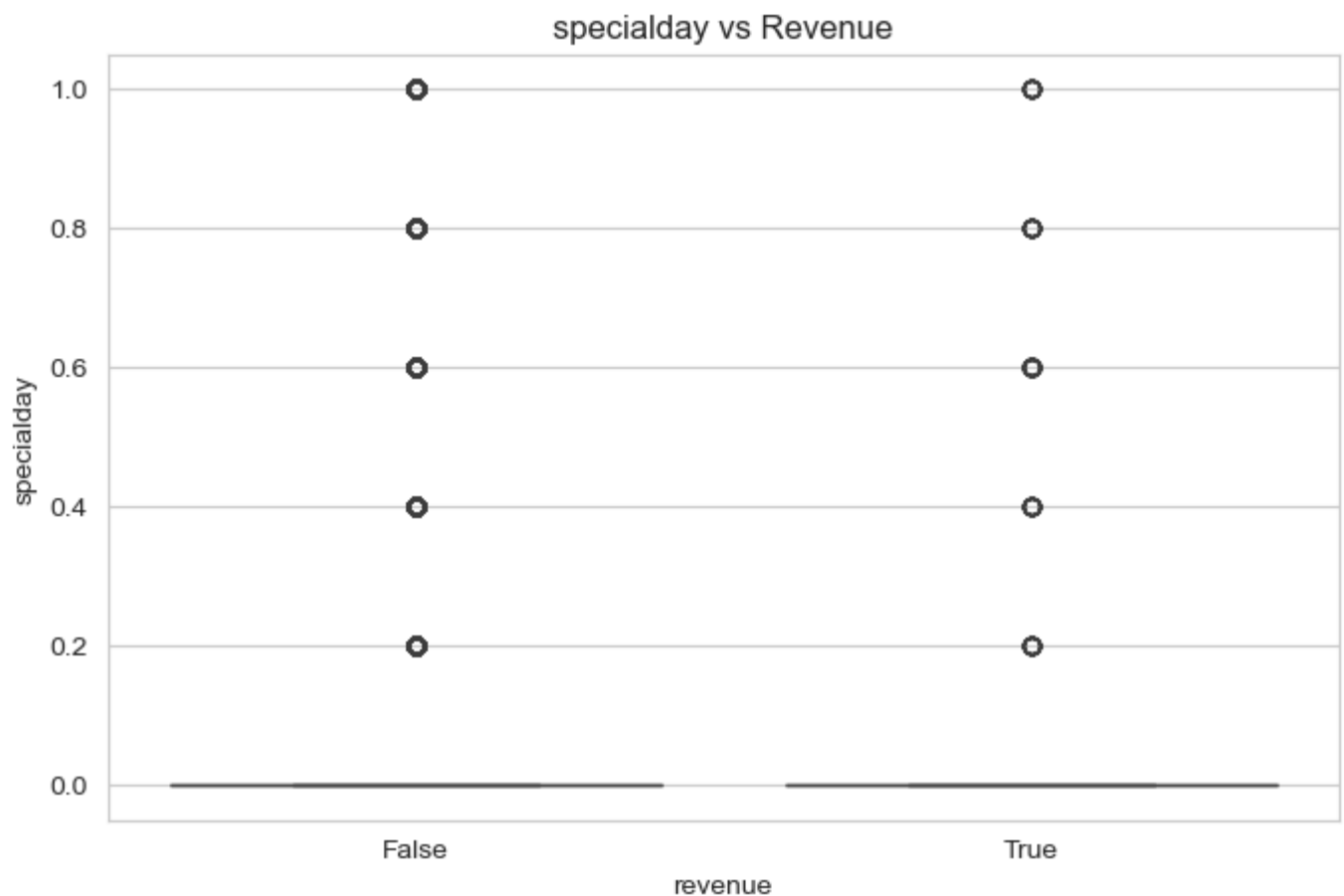
```
In [20]: for col in continuous_cols:
plt.figure(figsize=(8,5))
sns.boxplot(
    data=df,
    x='revenue',
    y=col
```

```
)  
plt.title(f'{col} vs Revenue')  
plt.show()
```









Observations

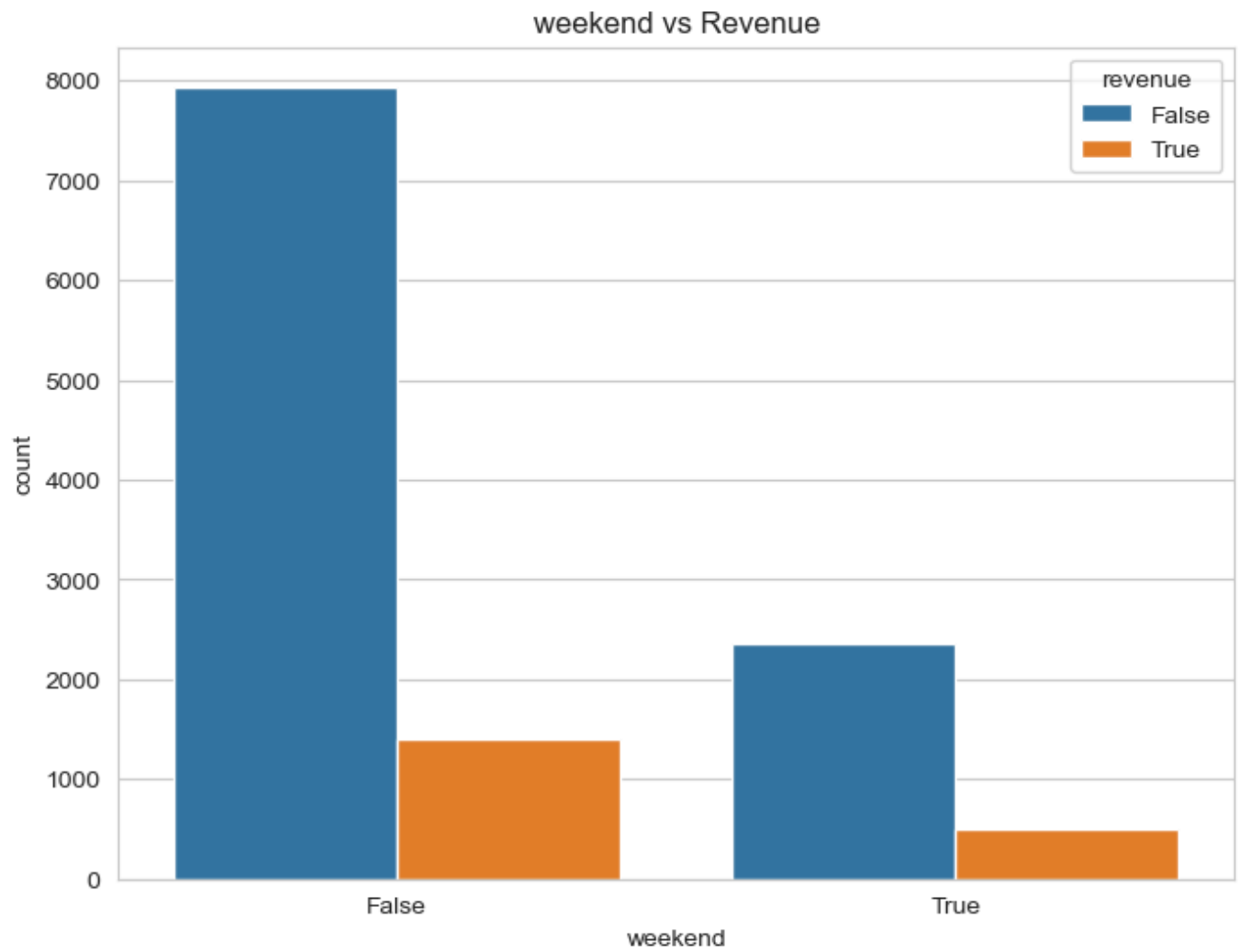
1. Administrative_Duration: Purchasing customers generally spend more time on administrative pages than non-purchasing customers.
2. Informational_Duration: Customers who make a purchase tend to spend slightly more time on informational pages.
3. ProductRelated_Duration: Purchasing customers spend significantly more time on product-related pages, indicating higher buying intent.
4. BounceRates: Customers who generate revenue have lower bounce rates, suggesting better engagement.
5. ExitRates: Lower exit rates are associated with customers who complete a purchase.
6. PageValues: Purchasing customers have considerably higher PageValues, making it one of the strongest indicators of revenue.
7. SpecialDay: Most sessions occur when SpecialDay = 0, and no clear relationship with revenue is observed.

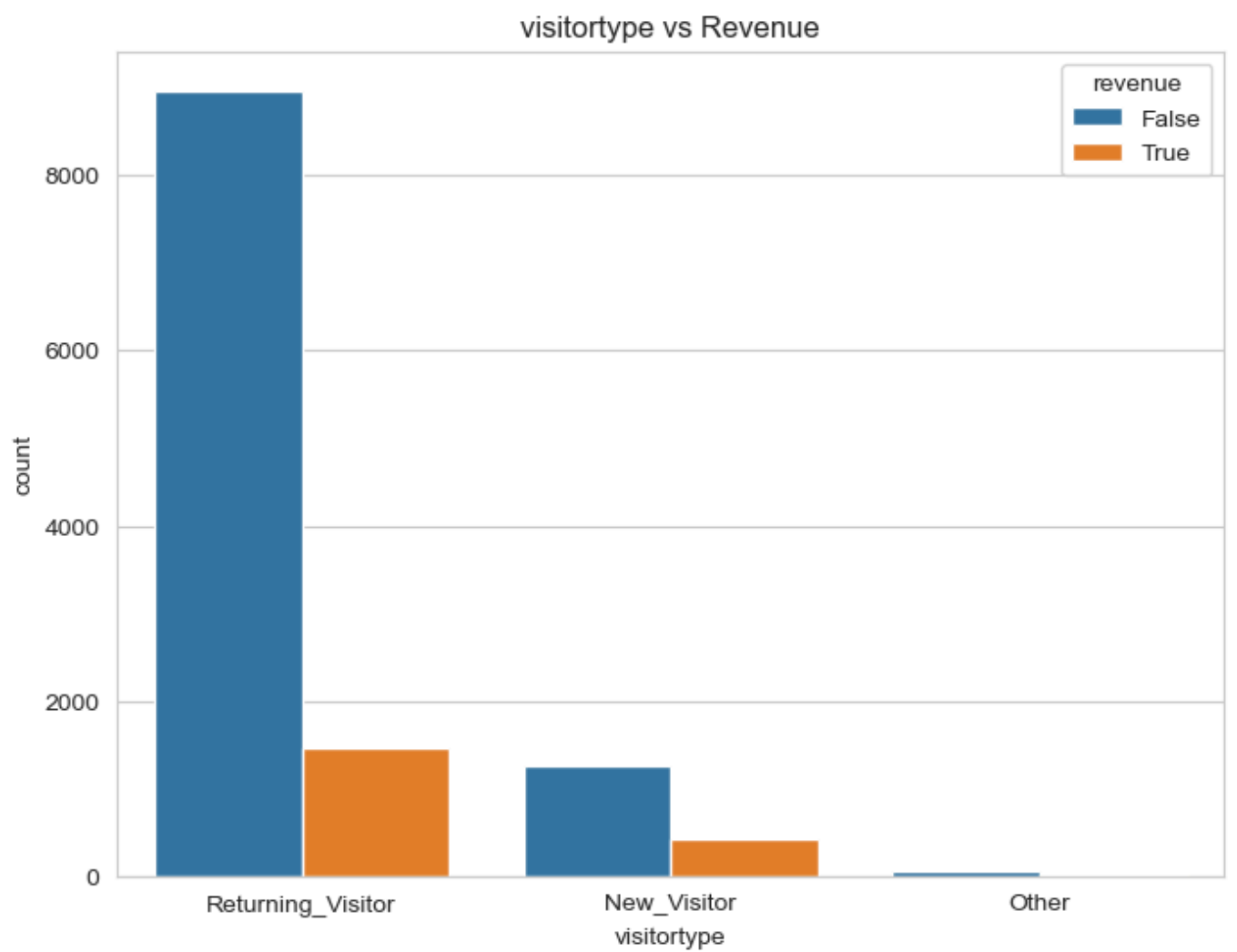
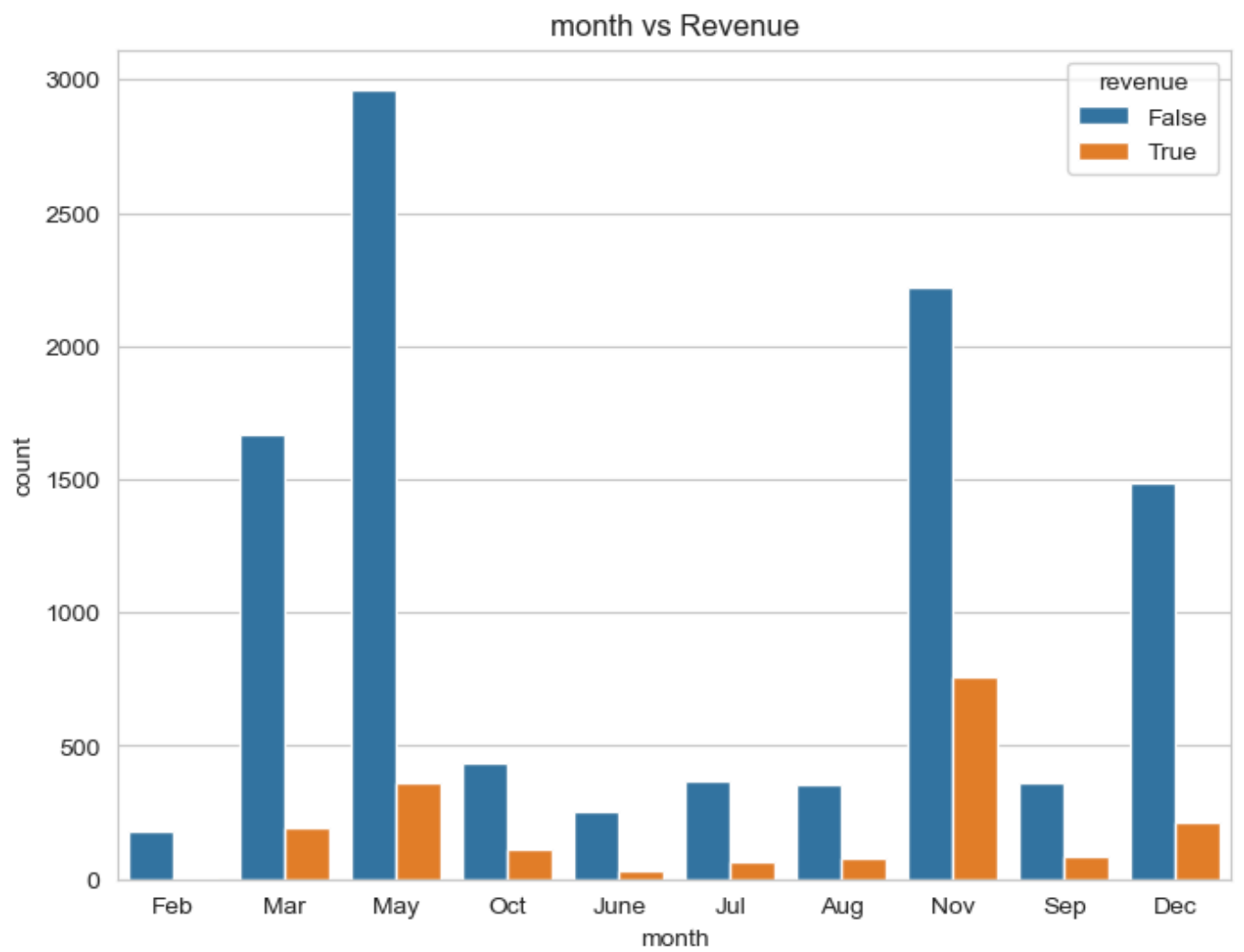
Bivariate analysis : Categorical Columns vs Revenue

```
In [21]: categorical_cols = ['weekend', 'month', 'visitortype']
for cols in categorical_cols:
    plt.figure(figsize=(8,6))
    sns.countplot(
        data=df,
        x=cols,
```



```
    hue='revenue'  
)  
plt.title(f'{cols} vs Revenue')  
plt.show()
```

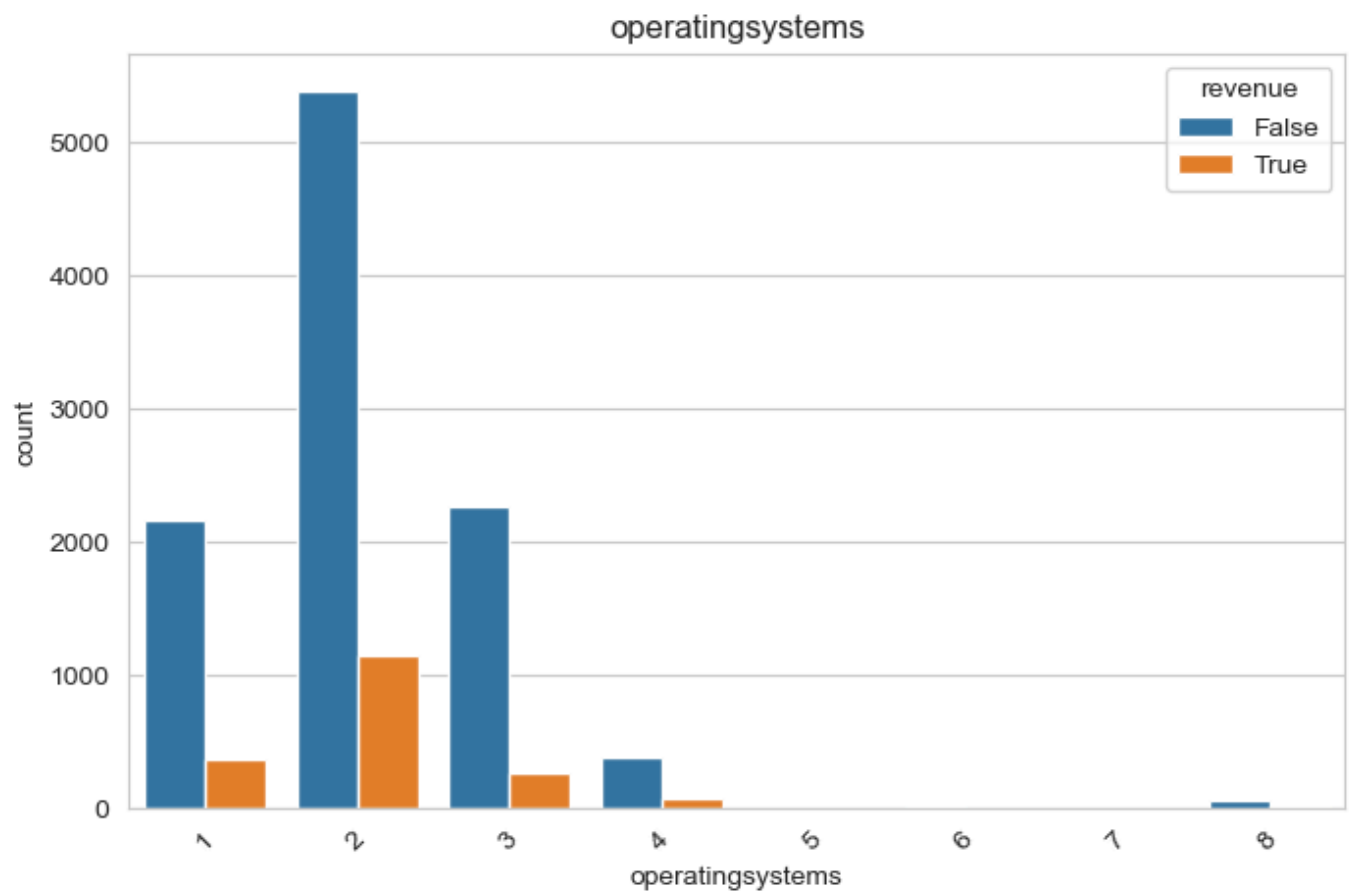




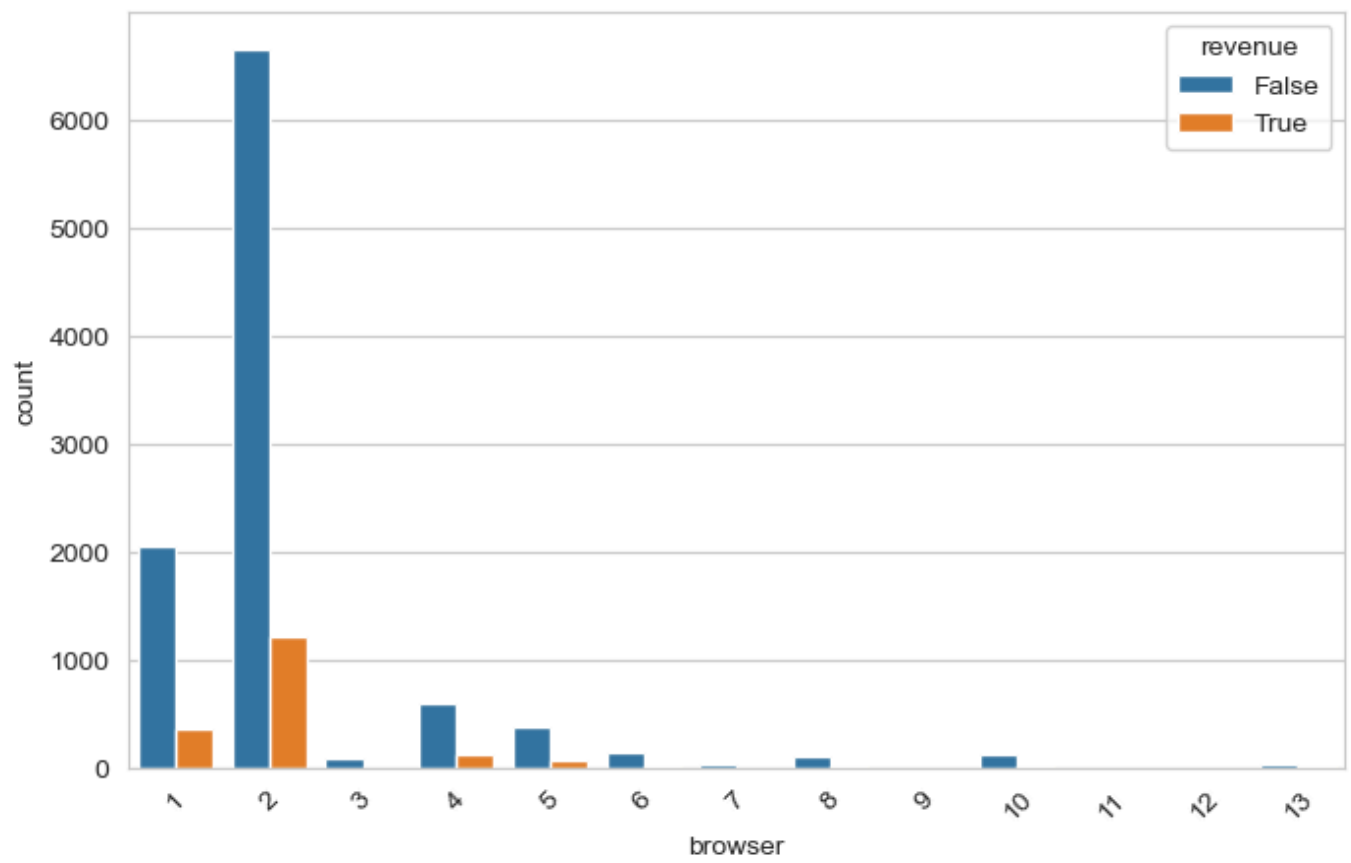
Observations:

1. Weekend: Non Weekend days has more revenue generations as compared to Weekend days
2. Month : Highest revenue generations comes from the month of 'NOVERMBER' , 'DECEMBER' and 'MAY'.
3. Visitor Type : Returning Visitors generate more revenue as compared to other categories.

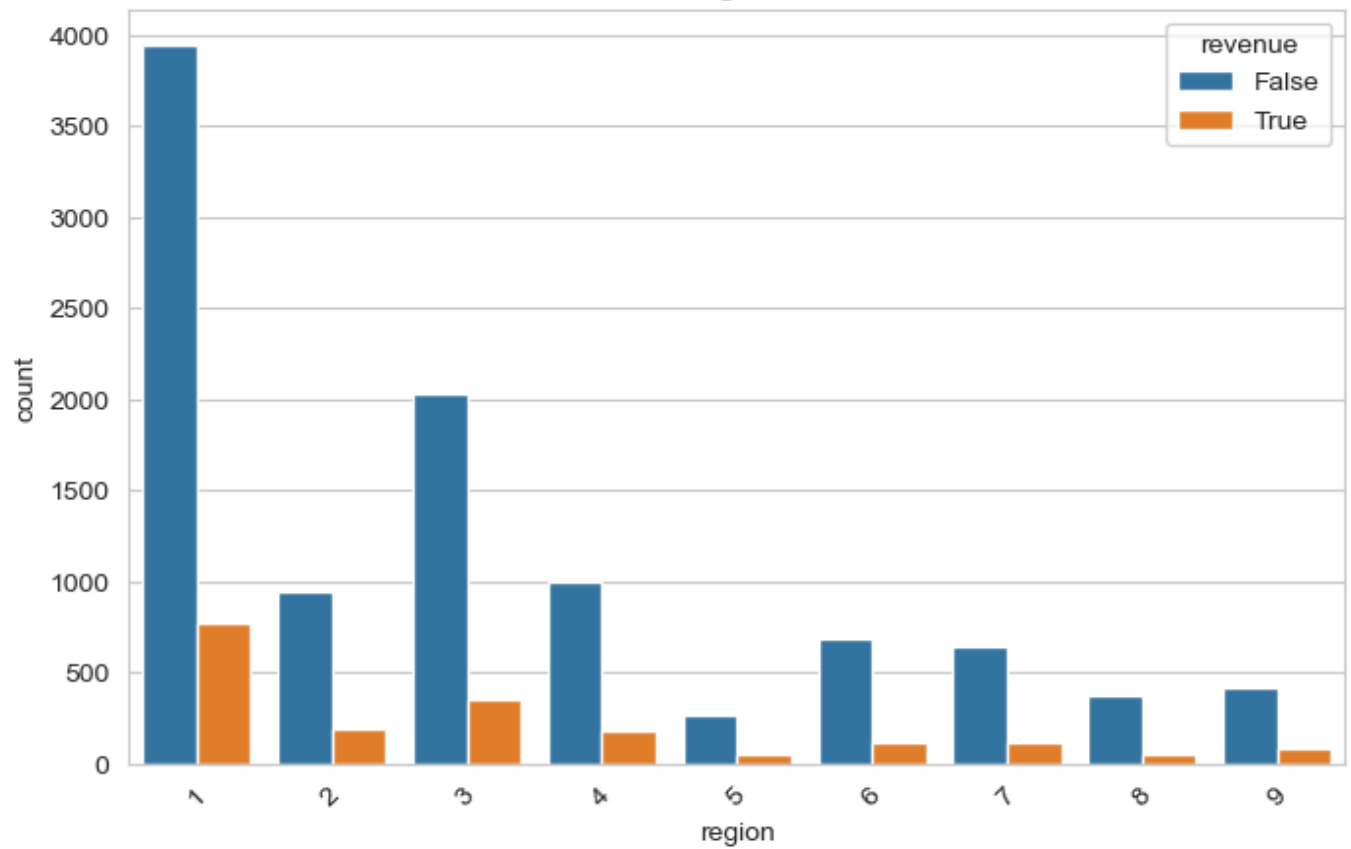
```
In [22]: categorical_int = [  
    'operatingsystems',  
    'browser',  
    'region',  
    'traffictype'  
]  
  
for col in categorical_int:  
    plt.figure(figsize=(8,5))  
    sns.countplot(  
        data=df,  
        x=col,  
        hue='revenue'  
    )  
    plt.title(col)  
    plt.xticks(rotation=45)  
    plt.show()
```

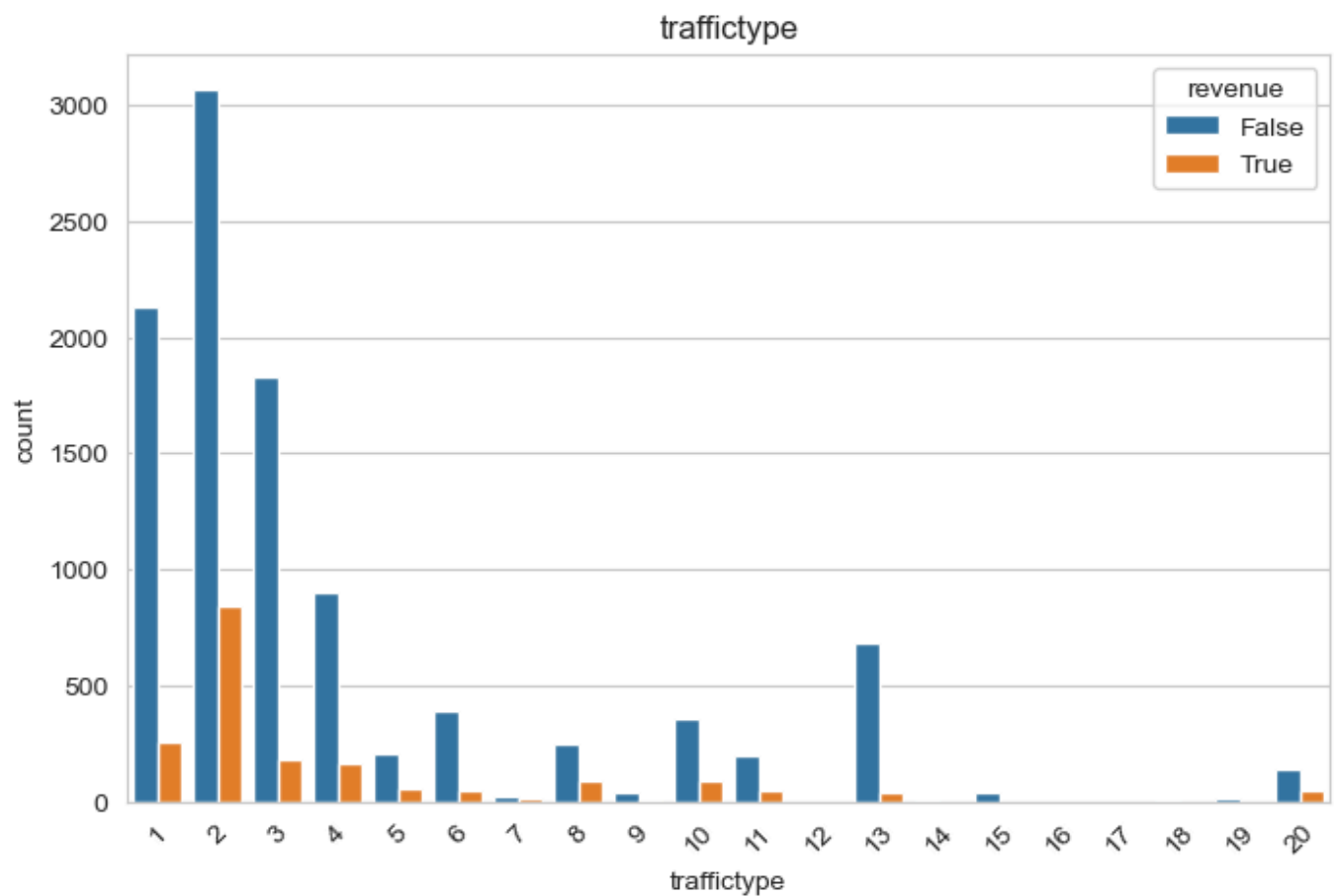


browser



region



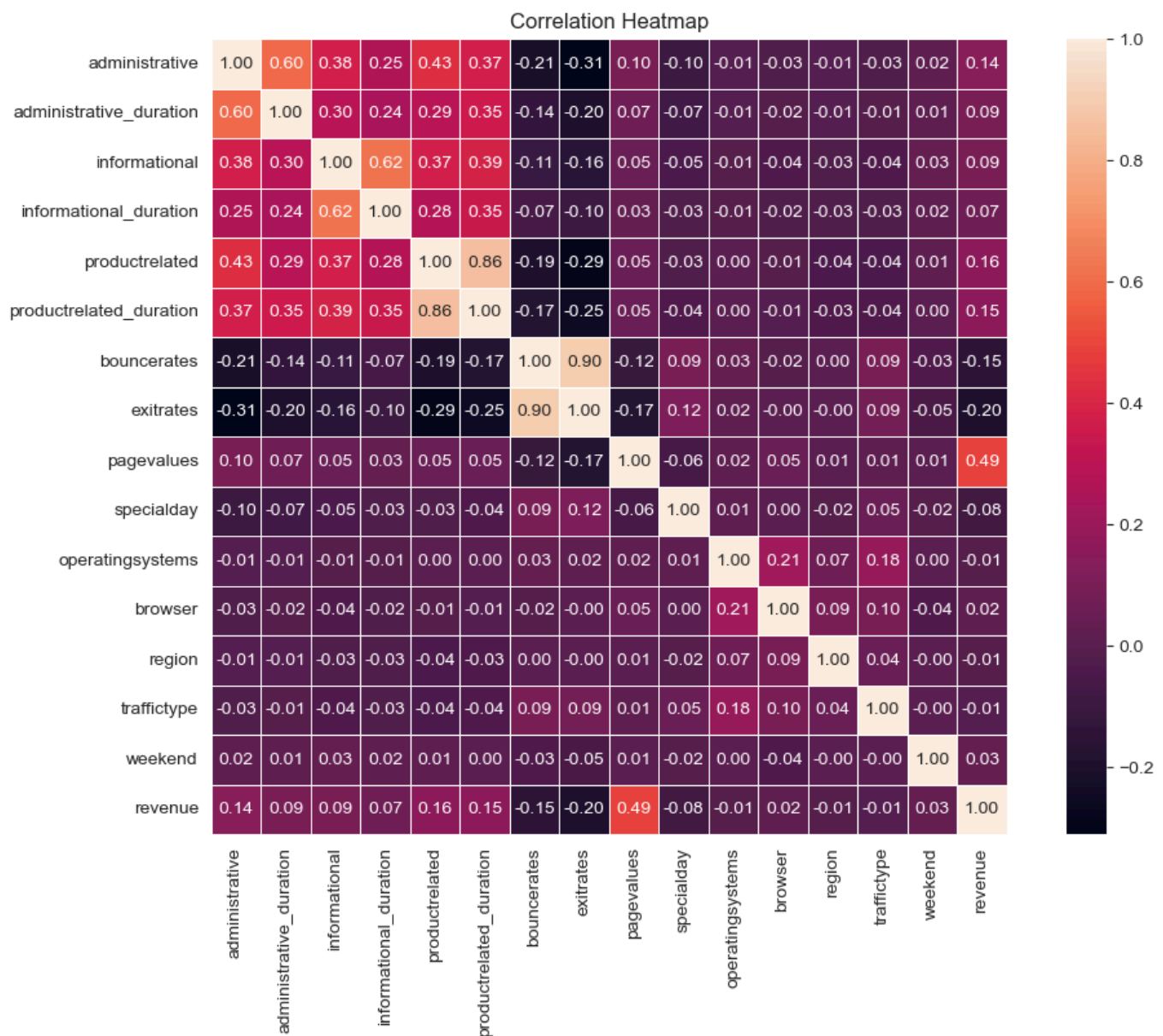


Observations:

1. OperatingSystems: Most website sessions and purchases originate from Operating System 2, while the remaining operating systems contribute relatively fewer sessions.
2. Browser: Browser 2 accounts for the highest number of website visits and purchases, whereas other browsers have comparatively lower user activity.
3. Region: Region 1 has the highest website traffic and purchases, although purchases are observed across all regions.
4. TrafficType: Traffic Types 2, 1, and 3 generate the highest number of visitors and purchases, while the remaining traffic sources contribute relatively less traffic.
5. Overall Observation User activity and purchases are concentrated in a few operating systems, browsers, regions, and traffic sources, indicating that these features may provide useful information for predicting customer purchase behavior.

```
In [23]: df['weekend'] = df['weekend'].astype(int)
df['revenue'] = df['revenue'].astype(int)
```

```
In [29]: plt.figure(figsize=(12,8))
corr = df.corr(numeric_only=True)
sns.heatmap(
    corr,
    annot=True,
    fmt='.2f',
    linewidth=0.5,
    square=True
)
plt.title("Correlation Heatmap")
plt.show()
```



Observations:

1. Most features exhibit weak to moderate correlations, indicating that multicollinearity is generally not a major concern.
2. ProductRelated and ProductRelated_Duration have a strong positive correlation (0.86), suggesting that users who visit more product pages also tend to spend more time on them.
3. BounceRates and ExitRates show a very strong positive correlation (0.90), indicating that both features capture similar aspects of user engagement.
4. PageValues has the highest positive correlation with Revenue (0.49), making it one of the strongest predictors of customer purchases.
5. BounceRates (-0.15) and ExitRates (-0.20) are negatively correlated with Revenue, suggesting that higher bounce and exit rates reduce the likelihood of a purchase.
6. ProductRelated (0.16), ProductRelated_Duration (0.15), and Administrative (0.14) have a weak positive correlation with Revenue, indicating that greater engagement with the website slightly increases purchase probability.
7. SpecialDay, OperatingSystems, Browser, Region, TrafficType, and Weekend show very weak correlations with Revenue, implying they have limited individual predictive power.

```
In [30]: df.to_csv("online_shoppers_cleaned.csv", index=False)
```

```
In [ ]:
```